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Top-down feedback matters: Functional impact of brainlike connectivity motifs on audiovisual integration

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eLife Assessment

This **valuable** study investigates the computational role of top-down feedback - a property found in biological circuits - in artificial neural network (ANN) models of the neocortex. Using hierarchical recurrent ANNs in an audiovisual integration task, the authors show that an anatomically inspired feedback motif induces a stable visual bias consistent with human perception and yields modest but meaningful benefits for learning dynamics and robustness. The strength of evidence is **solid**: the modeling, analyses, and controls mostly support the central claim that top-down feedback motifs impose persistent inductive biases that shape functional specialization and behavior. But the evidence for a broad, general framework that predicts behavior remains only partially supported, and the Methods would benefit from a compact, reproducible summary of hyperparameters and architectural details.

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Abstract

Artificial neural networks (ANNs) are an important tool for studying neural computation, but many features of the brain are not captured by standard ANN architectures. One notable missing feature in most ANN models is top-down feedback, i.e. projections from higher-order layers to lower-order layers in the network. Top-down feedback is ubiquitous in the brain, and it has a unique modulatory impact on activity in neocortical pyramidal neurons. However, we still do not understand its computational role. Here we develop a deep neural network model that captures the core functional properties of top-down feedback in the neocortex, allowing us to construct hierarchical recurrent ANN models that more closely reflect the architecture of the brain. We use this to explore the impact of different hierarchical recurrent architectures on an audiovisual integration task. We find that certain hierarchies, namely those that mimic the architecture of the human brain, impart ANN models with a light visual bias similar to that seen in humans. This bias does not impair performance on the audiovisual tasks. The results further suggest that different configurations of top-down feedback make otherwise identically connected models functionally distinct from each other, and from traditional feedforward and laterally recurrent models. Altogether our findings demonstrate that modulatory top-down feedback is a computationally relevant feature of biological brains, and that incorporating it into ANNs affects their behavior and constrains the solutions it's likely to discover.

1 Introduction

Artificial neural networks (ANNs) often draw inspiration from the brain and serve in turn as tools with which to model it (Doerig et al., 2023 [DOI](#)). For example, deep convolutional neural networks (CNN)—originally inspired by hierarchical feature detection in the mammalian brain (Fukushima, 1980 [DOI](#))—predict behavior and representations in visual cortex with high accuracy (Cadena et al., 2019 [DOI](#); Yamins & DiCarlo, 2016 [DOI](#)). However, there are still significant gaps in most deep neural networks’ ability to capture behavior, particularly when presented with ambiguous, challenging stimuli (Geirhos et al., 2018 [DOI](#); Kar et al., 2019 [DOI](#)). This gap is thought to be a result, in part, of the exclusively feedforward structure of CNNs, as opposed to the biological brain, which consists of feedforward as well as local and top-down recurrent connections (Kar et al., 2019 [DOI](#); van Bergen & Kriegeskorte, 2020 [DOI](#)). Local recurrence is modeled in a subfamily of ANNs known as recurrent neural networks (RNNs), and there have been studies using RNNs to model sensory processes in the brain (Kubilius et al., 2018 [DOI](#)). In contrast, top-down feedback has largely been neglected in deep models, with a few exceptions (Islah et al., 2023 [DOI](#); Naumann et al., 2022 [DOI](#); Tsai et al., 2024 [DOI](#); Wybo et al., 2022 [DOI](#)). It’s unclear whether omitting top-down feedback changes the behavior of a model or has an impact on processing in a deep model.

As such, we built ANNs with biologically-inspired top-down feedback and examined whether different configurations of feedback vs feedforward connectivity had any relevant effect on model behavior and preferences. Importantly, top-down feedback connections are functionally and physiologically distinct from feedforward connections. They typically connect higher order association areas to lower order sensory areas (Felleman & Van Essen, 1991 [DOI](#)). Data suggests that top-down inputs in the neocortex are modulatory (Larkum, 2004 [DOI](#); Sherman & Guillery, 1998 [DOI](#)), meaning they alter the magnitude of neural activity, but do not drive it themselves. It would be beneficial for computational neuroscience if ANN models allowed us to explore the potential unique impacts of top-down feedback on computation in the neocortex.

For this study, we chose to examine the effects of ANNs with top-down feedback on an audiovisual categorization task. We chose audiovisual tasks because feedback connections can also span sensory modalities, and the auditory areas (which project directly to V1 in primates (Clavagnier et al., 2004 [DOI](#))) are speculated to sit higher on the sensory-association hierarchy than primary visual areas (King & Nelken, 2009 [DOI](#)). As such, audio modulation of visual processing likely occurs via top-down feedback across multiple cortical sites. Audiovisual interplay produces many well-known functional effects, such as the McGurck effect (McGurck & Macdonald, 1976 [DOI](#)) and sound-induced flash illusions (Shams et al., 2000 [DOI](#)), making it a particularly interesting testbed for studies of the computational role of top-down feedback.

To study the effect of top-down feedback on such tasks, we built a freely available code base for creating deep neural networks with an algorithmic approximation of top-down feedback. Specifically, top-down feedback was designed to modulate ongoing activity in recurrent, convolutional neural networks. We explored different architectural configurations of connectivity, including a configuration based on the human brain, where all visual areas send feedforward inputs to, and receive top-down feedback from, the auditory areas. The human brain-based model performed well on all audiovisual tasks, but displayed a unique and persistent visual bias compared to models with only driving connectivity and models with different hierarchies. This qualitatively matches the reported visual bias of humans engaged in audio-visual tasks (Posner et al., 1976 [DOI](#); Stokes & Biggs, 2014 [DOI](#)). Our results confirm that distinct configurations of feedforward/feedback connectivity have an important functional impact on a model’s behavior. Therefore, top-down feedback captures behaviors and perceptual preferences that do not manifest reliably in feedforward-only networks. Further experiments are needed to clarify whether top-down feedback helps an ANN fit better to neural data, but the results show that top-down feedback affects the processing of stimuli and is thus a relevant feature that should be considered for deep ANN models in computational neuroscience more broadly.

2 Results

2.1 Framework for modeling top-down feedback with RNNs

Based on neurophysiological data, we implemented a deep neural network model where each layer corresponds to a brain region and receives two different types of input: feedforward and feedback. Feedforward input drives the activity of the layer as it does in regular ANNs. In contrast, inspired by the physiology of apical dendrites in pyramidal neurons, feedback input to each neuron is integrated and then run through another non-linear function. The result of this is then multiplied with the integrated feedforward activation (Fig. 1A [↗](#), left). As such, feedback input cannot activate a neuron that is not receiving feedforward input, nor can it decrease the activity of the neuron, but it can modulate its level of activity (Fig. 1A [↗](#), right). This mimics the modulatory role of top-down feedback in the neocortex as observed in neurophysiological experiments (Larkum, 2004 [↗](#); McAdams & Maunsell, 1999 [↗](#)). But, we note that experimental and modeling research suggests that top-down feedback can sometimes be weakly driving (Larkum, 2004 [↗](#); Reynolds et al., 2000 [↗](#); Shai et al., 2015 [↗](#)), i.e. it can reduce the threshold for activation for the neuron, a form of feedback referred to as “composite” feedback (Shai et al., 2015 [↗](#)). We explore the impact of both purely multiplicative feedback and composite feedback below.

We implemented retinotopic and tonotopic organization, as well as local recurrence, by using a Convolutional Gated Recurrent Unit (GRU) to model each brain region (Ballas et al., 2016 [↗](#); Cho et al., 2014 [↗](#)). These models were end-to-end differentiable, and therefore, we could train them with backpropagation-of-error. Importantly, there is nothing about this modeling framework that demands a specific type of training, meaning that the models created within this framework can undergo supervised, unsupervised, or reinforcement learning, though here we used supervised tasks (see Methods).

For the audiovisual categorization tasks, we constructed a series of ANNs with four layers corresponding to ventral visual regions (V1, V2, V4, and IT) and three layers corresponding to ventral auditory regions (A1, Belt, A4). For one ANN, which we will call the “brainlike” model, we used the externopyramidalisation of each cortical area as seen in human histological data to determine the hierarchical position of said area in the network (Fig. 1B [↗](#)). Externopyramidalisation refers to the relative differentiation of the supragranular layers in the cortex (Sanides, 1962 [↗](#)). Percentage of supragranular projecting neurons (SLN) is a longstanding measure of hierarchical distance in mammals, based on experimental observations that long-range feedforward connections originate primarily from the supragranular layers, while feedback connections are infragranular in origin (Barone et al., 2000 [↗](#); Felleman & Van Essen, 1991 [↗](#); Markov & Kennedy, 2013 [↗](#)). Externopyramidalisation is an indirect estimate of supragranular projecting neurons, based on observations in primates that feedforward-projecting sensory areas feature dense, highly differentiated supragranular layers (Gerbella et al., 2007 [↗](#)) while feedback-projecting transmodal areas have less differentiated supragranular layers and a larger infragranular layer (Morecraft et al., 2012 [↗](#)). Researchers thus assume that higher externopyramidalisation scores (i.e. thicker and more differentiated supragranular layers) indicates a “lower order” region that sends more feedforward connections (Goulas et al., 2018 [↗](#)). We make the same assumption in-line with previous literature quantifying hierarchical distance from human histological data (Paquola et al., 2019 [↗](#); Saberi et al., 2023 [↗](#)), but we note that our modeling framework could easily use other metrics for determining position in the cortical hierarchy (Paquola et al., 2020 [↗](#); Wagstyl et al., 2015 [↗](#); Zilles & Amunts, 2009 [↗](#)).

This resulted in an architecture wherein visual areas provided feedforward input to all auditory areas, whereas auditory areas provided feedback to visual areas (Fig. 1C [↗](#), left), similar to an existing hypothesis on primate audiovisual processing (King & Nelken, 2009 [↗](#)). In another ANN, that we call the “reversed” model, the directionality was reversed, i.e. all auditory areas feed forward to the visual areas whereas visual areas provide feedback to auditory areas. In this model, directionality of connections within a modality were kept the same as the brain-based model (Fig. 1C [↗](#), middle). In three additional control ANNs, the directionality of connections between each pair of regions was determined by a coin toss (Fig. 1C [↗](#), right).

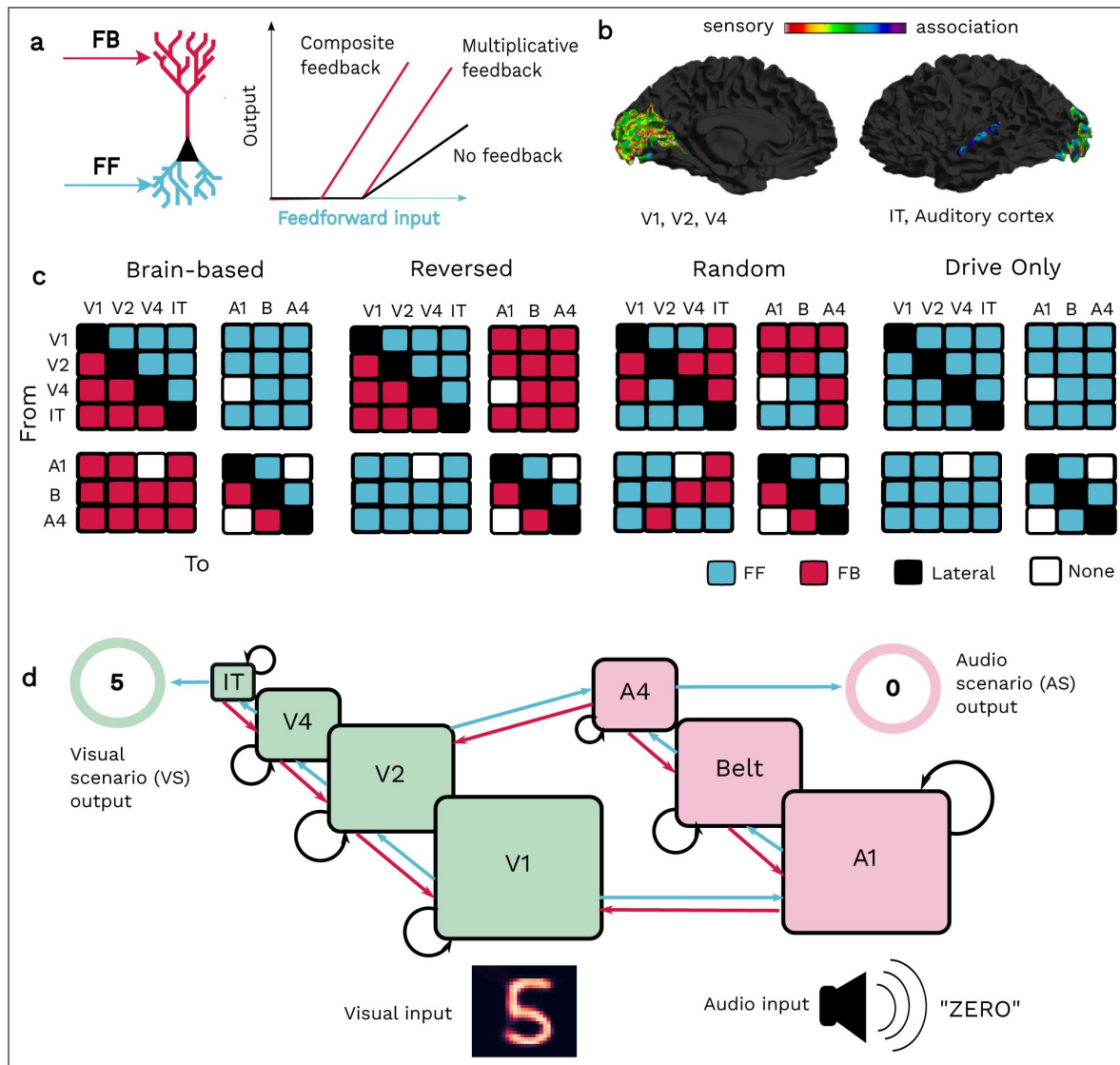


Figure 1. Description of models.

a, Each area of the model receives driving feedforward input and modulatory feedback input. Feedback input alters the gain of a neuron, but it doesn't affect its threshold of activation (multiplicative feedback). In later experiments, we explore an alternative mechanism, where it weakly affects the threshold of activation (composite feedback). **b**, Modeled regions and their externopyramidalisation values (i.e thickness and relative differentiation of supragranular layers, used as proxy measure for sensory-associational hierarchical position). Note higher overall externopyramidalisation values in the occipital lobe compared to the temporal lobe. **c**, Using above hierarchical measures, we constructed models where each connection has a direction (i.e regions send either feedforward or feedback connections to other regions). In the brainlike model based on human cytoarchitectural data, where all visual regions send feedforward to and receive feedback from the auditory regions. In the reverse model, all auditory regions send feedforward to and receive feedback from visual regions, while connections within a modality remain the same. **d**, The resulting ANN. Outputs of image identification tasks are read out taken from IT, outputs of audio identification tasks are read out from A4, an auditory associational area. Connections between modules are simplified for illustration.

Altogether, the framework we developed allowed us to explore the functional implications of top-down feedback in multi-layer RNNs. As well, this framework allowed us to examine the functional impact of a RNN architecture inspired by the human brain on audiovisual processing, and compare it to other non-brainlike architectures.

2.2 Effect of top-down feedback on visual tasks with auditory cues

In order to examine the functional impact of different top-down feedback architectures, we first trained multiplicative feedback models (i.e. not composite feedback) on an image categorization task with additional auditory stimuli. The task for the ANN was to correctly identify the category based on the image, but the images were sometimes ambiguous (which were created using a variational autoencoder, see Methods). In cases where the visual image was ambiguous, the network was provided with a disambiguating auditory cue (we refer to this as Visual-dominant Stimulus case 1, or VS1; Fig. 2a [↗](#), top). In addition, we presented the model with situations where the visual input was unambiguous and the auditory input was misleading (VS2; Fig. 2a [↗](#), bottom), in order to test whether the models could learn to ignore misleading audio inputs. Crucially, the models were never told which stimulus scenario they were encountering. As such, they had to learn to rely on the auditory inputs when the visual inputs were ambiguous (VS1), and ignore the auditory inputs when the visual inputs were unambiguous (VS2). Models were trained in mini-batches on both VS1 and VS2 stimuli, then tested on held out data (see Methods for details).

In stimulus condition VS1, all of the models were able to learn to use the auditory clues to disambiguate the images (Fig. 2b [↗](#)). However, the brainlike model learned to rely on auditory inputs more slowly, whereas the reversed model was exceptionally good at integrating auditory information. In comparison, in VS2, we found that the brainlike model learned to ignore distracting audio inputs quickly and consistently compared to the random models, and a bit more rapidly than the auditory information (Fig. 2c [↗](#)). These data show that the brainlike model is biased towards visual stimuli compared to the other models.

We next wanted to probe this visual bias further by examining how the models relied on either visual or auditory inputs when neither was unambiguously relevant to the task. Thus, in another set of test stimuli, we examined what happened when the models were presented with non-matching auditory input and ambiguous visual input (VS3; Fig. 2d [↗](#)). Notably, VS3 was not given as a task during training; the models had been trained on stimuli from the VS1 and VS2 conditions. In this situation, we were interested in whether the models were more likely to align their answers to the visual or auditory stimuli, given that neither provided a clear and correct answer. We found that the brainlike model aligned with the visual input from the start, while all other models had to learn to do so further along in training (Fig. 2e [↗](#)). Importantly, if we trained the models purely on unambiguous images (VS4; Fig. 2f [↗](#)), all of the models were equally adept at learning (Fig. 2g [↗](#)). As such, the visual bias shown by the brainlike model did not reflect a more general increased capability with visual inputs, rather, it reflected an inductive bias in audio-visual integration created by the top-down feedback architecture. Altogether, our results demonstrated that both the architecture of top-down feedback can impact audio-visual processing, and that the specific architecture inspired by the human brain has a visual bias in these tasks.

2.3 Effect of top-down feedback on auditory tasks with visual cues

To see if the visual bias persisted for a non-visual task, we then trained the models from scratch on an audio recognition task, where the ANNs now had to identify ambiguous sounds using visual stimuli as clues (AS1; Fig. 3a [↗](#), top) and ignore distracting images when they're not needed (AS2; Fig. 3a [↗](#), bottom). As before, the models were trained on data from the AS1 and AS2 conditions, then tested on held out data points. Notably, only the brainlike model and one of the random models learned to use the visual stimuli at all, across 10 random seeds (Fig. 3b [↗](#)). In contrast, all of the models learned to ignore the distractor image (Fig. 3c [↗](#)). This indicates that the brainlike model is more inclined to use visual stimuli for the task than the other models.

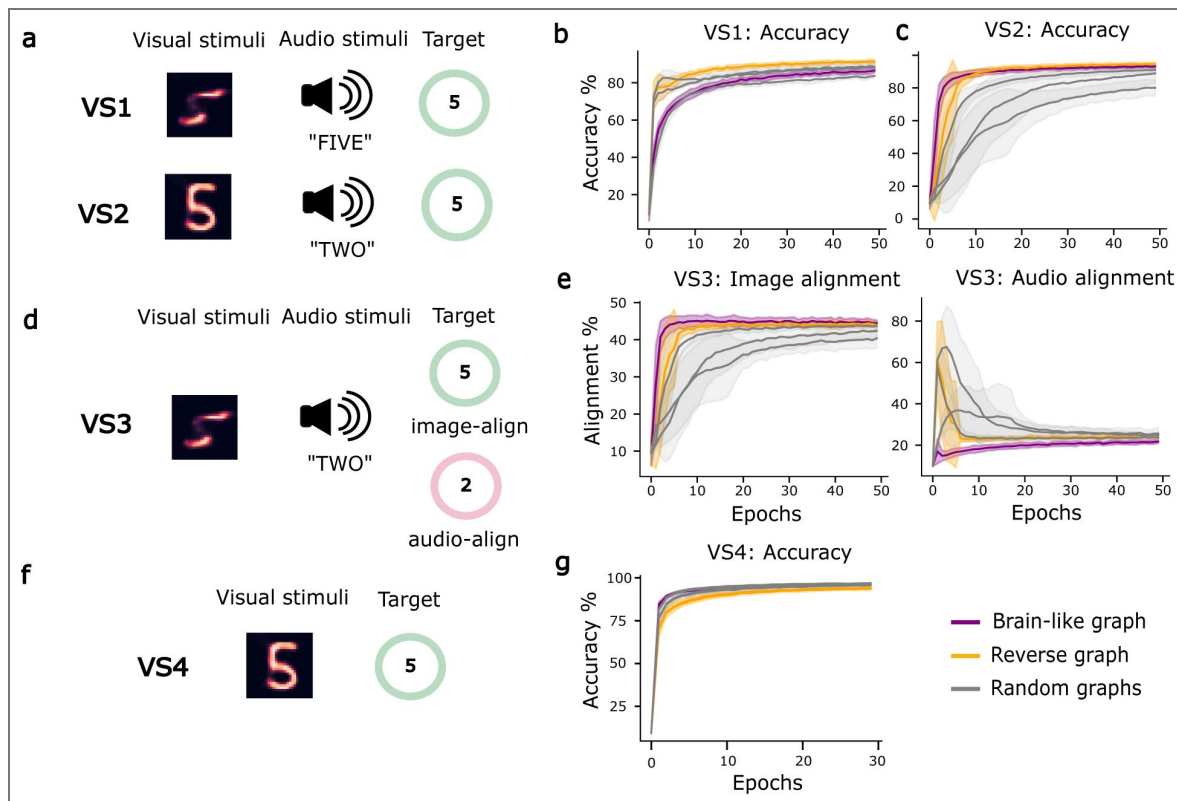


Figure 2. Multimodal visual tasks.

a, Training conditions. Models must identify the visual stimulus given an ambiguous image and a matching audio clue (VS1) or unambiguous image and distracting audio (VS2). **b-c**, Accuracy across epochs for tasks VS1 and VS2 on holdout datasets, **d** Trained models were given an ambiguous visual stimulus and a nonmatching audio stimulus (VS3) to assess which modality they align most closely with. **e**, Alignment of trained models across epochs based on task VS3. Scenario was shown to model at the end of each training epoch, but never during training. **f**, Models were additionally trained and tested on image stimuli only to assess their baseline performance (VS4). **g**, Accuracy of models across epochs on task VS4

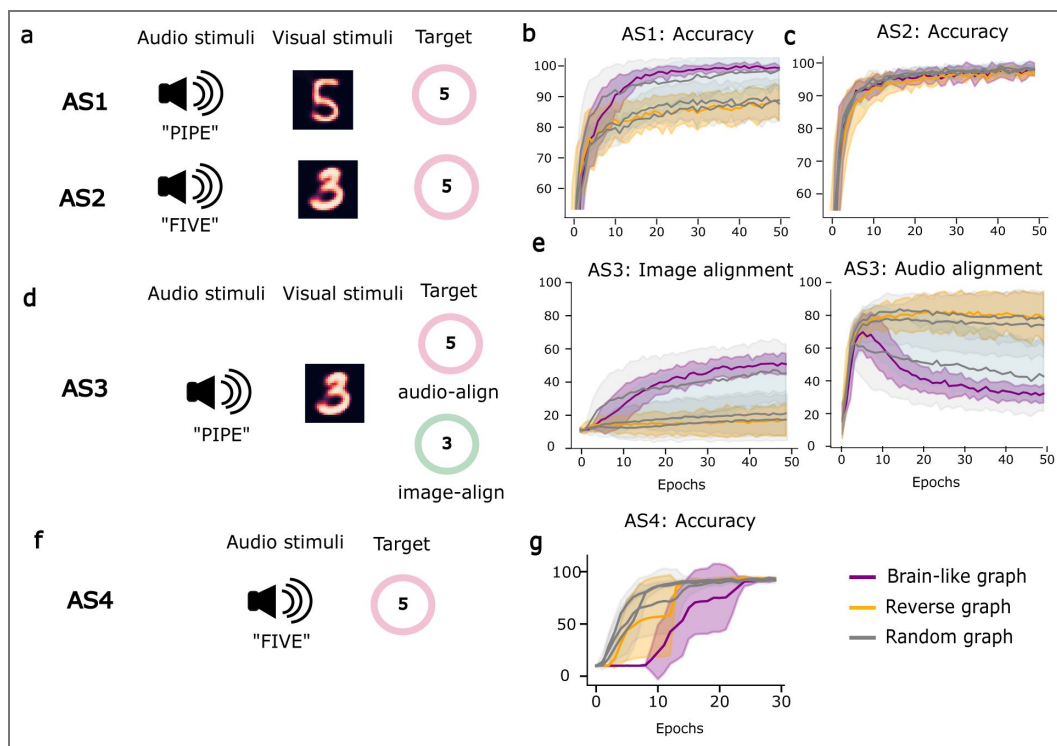


Figure 3. Multimodal auditory tasks.

a, Training conditions. Models must identify the auditory stimulus given an ambiguous audio and a matching visual clue (AS1) or unambiguous audio and distracting image (AS2). **b-c**, Accuracy across epochs for tasks AS1 and AS2 on holdout datasets, **d**, Trained models were given an ambiguous audio stimulus and a nonmatching visual stimulus (AS3) to assess which modality they align most heavily with. **e**, Alignment of trained models across epochs based on task AS3. Scenario was shown to model at the end of each training epoch, but never during training. **f**, Models were additionally trained and tested on audio stimuli only to assess their baseline performance (AS4), **g**, Accuracy of models across epochs on task AS4

Next, we examined how the models responded when neither input was unambiguously informative. In a test scenario with non-matching visual stimuli and ambiguous audio (AS3; Fig. 3d), the brainlike model aligned more heavily with the visual input than most other models (Fig. 3e). Thus, despite the task being primarily auditory, the brainlike model retained its clear visual bias. Interestingly, unlike in the case of visual tasks, there was a difference between the models in terms of their ability to learn purely from unambiguous auditory stimuli (AS4; Fig. 3f). While all models could achieve the same final error rate, the brainlike model took longer to learn from purely auditory inputs than the other models. Our results suggest that the architecture of top-down feedback in the human brain may provide an inductive bias that favours visual inputs over auditory inputs.

2.4 Effects of composite top-down feedback

As noted above, neurophysiological evidence suggests that in addition to modulating activity in a multiplicative manner, top-down feedback in the neocortex can be weakly driving as well, shifting not just the gain but also the threshold of activation for a neuron (Fig. 1A) (Shai et al., 2015). As such, we performed the same experiments as above, but now with composite feedback, i.e. feedback that has both a multiplicative and an additive effect. Notably, composite feedback is more akin to a feedforward connection (due to the weakly driving additive component). Therefore, in order to isolate for the impact of composite top-down feedback we added a control of an identically connected network consisting only of driving feedforward connections (i.e. no top-down feedback anywhere), which we called the drive-only model. We then compared the brainlike composite feedback models to the original controls and the new drive-only control.

Interestingly, we found that composite feedback improved the performance of all models on tasks that they previously had difficulty with, while maintaining performance on the tasks they were already strong on (Fig. 4a-d). Specifically, in the visual task with ambiguous visual stimuli and informative audio cues (VS1), we observed faster learning and massively improved final accuracy in the brainlike and random models, which lagged behind the reverse model when using multiplicative feedback (Fig. 4a). Conversely, the reverse model with multiplicative feedback had difficulty using visual clues when identifying ambiguous audio (AS1). Composite feedback improved the performance of the reverse model on this task, but the already high-performing brainlike model remained unaffected by the change (Fig. 4b). The effect may be due to the slightly increased size of the composite models, as we observed that bigger models also receive a similar boost in AS1 performance (Fig. S2a). However, the composite models are 1K parameters larger than the models with multiplicative feedback (compared to an overall 280K) while the models in Fig. S2 are roughly three times the size.

The impact of composite feedback was more mixed in the situation where the visual input was unambiguous and the auditory input was misleading (VS2). The brainlike model and most random models showed better final accuracy (Fig. 4c), but a few random models only performed well in VS1 and not VS2 (Fig. 4c, rightmost outliers). The only task where composite feedback had no noticeable effect was when the auditory input was unambiguous and the visual clue was misleading (AS2), likely because all models already performed at a high level with multiplicative feedback (Fig. 4d). Interestingly, drive-only models (i.e. feedforward-only interlayer connections and lateral recurrence) consistently performed at an accuracy roughly equivalent to that of the brainlike composite feedback model (Fig. 4a-d, blue lines). These data imply that composite feedback may be a more general purpose computational mechanism that can be used for efficient multi-modal learning with the correct architecture, but based on accuracy alone, it did not behave noticeably differently from a drive-only model.

We next wondered whether the difference between composite top-down feedback and the drive-only model could be observed in terms of the visual bias of the networks. Therefore, we examined the visual biases of the models in cases where no unambiguous correct input was given (VS3 and AS3). We found that in both the visual tasks (Fig. 4e) and the auditory tasks (Fig. 4f) the visual bias of the brainlike model was still present. In contrast, the drive-only model showed biases between that of brainlike and reverse models with a larger variability between seeds than the

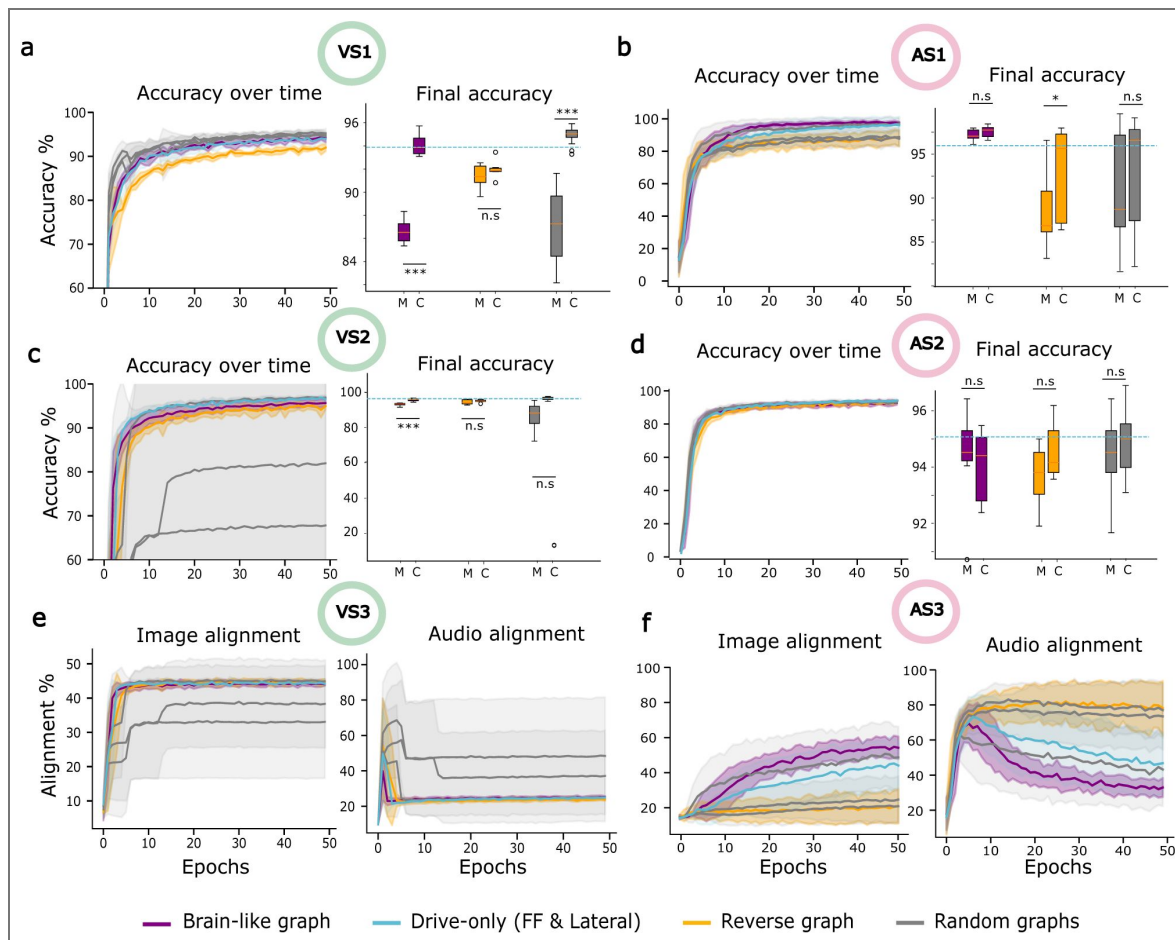


Figure 4. Composite versus multiplicative feedback in multimodal tasks.

a, c, e, Test performances of models with composite feedback and drive-only models trained on visual tasks (VS1 and VS2). The final epoch accuracy of models with composite feedback (C) is compared to that of models with multiplicative feedback (M) shown in previous figures **b, d, f,** Test performances of models trained on auditory tasks (AS1 and AS2).

brainlike model, suggesting the drive-only model privileged the visual and auditory input based on task demands and initialization. This shows that the inductive bias towards vision in the brainlike model depended on the presence of the multiplicative component of the feedback, and therefore, the visual bias is a function of the network architecture. This visual bias shows under all conditions for the network while the bias of the drive-only model varies based on task and initialization. Altogether, these results suggest that composite feedback as seen in the neocortex is an effective mechanism for multi-modal integration, and that the architecture of feedback in the human brain provides an inductive bias towards visual stimuli.

2.5 Effects of top-down feedback on task-switching

Our results so far showed that top-down feedback imparts a persistent inductive bias depending on the network architecture. However, top-down feedback is thought to be particularly important for context-dependent task switching (Li et al., 2004 [\[1\]](#); Liu et al., 2021 [\[2\]](#)). Thus, we wanted to know if the brainlike model's visual bias lent it any advantage over other networks, particularly in tasks that required flexible responses to context.

As such, we trained the models on all four training tasks (VS1, VS2, AS1, AS2) at the same time. To do so, we augmented all the models with a new output region that takes input from IT and A4. This new region in the model can be thought of as broadly representing higher order multimodal regions in the brain (Fig. 5a [\[3\]](#)). This multimodal region additionally received a binary attention flag with each pair of inputs indicating the target stimulus, i.e. whether to attend to the visual or auditory streams (Fig. 5b [\[4\]](#)). As in previous tasks, the models had to determine on their own whether the extraneous stimulus was useful for identifying the target stimulus (VS1, AS1) or a distraction to be ignored (VS2, AS2).

All models quickly learned to identify ambiguous visual stimuli using the auditory input (Fig. 5c [\[5\]](#), left), but the brainlike model learned to ignore distracting audio quicker than all other models including the drive-only model (Fig. 5c [\[5\]](#), right). Similarly, the brainlike model quickly learned to use visual clues when identifying ambiguous audio (Fig. 5d [\[6\]](#), left). However, it lagged behind the other models when it had to ignore the visual stimuli (Fig. 5d [\[6\]](#), right). When presented with two ambiguous stimuli of different labels, the brainlike model again aligned more frequently with the visual stimuli (Fig. 5e [\[7\]](#)). In contrast, the drive-only model started out more heavily aligned to the auditory stimuli and slowly aligned with the visual stimuli through training, suggesting it conforms to task demands and does not possess the same inductive bias for visual tasks (Fig. 5e [\[7\]](#)). Interestingly, the reverse model and two random models struggled to learn to ignore auditory stimuli (Fig. 5b [\[4\]](#)). This is likely because the auditory dataset is smaller, less variable and thus easier to learn than the visual dataset, causing certain models to shortcut their training and rely excessively on the audio inputs (see also Fig. 3b [\[8\]](#) and Fig. 3e [\[9\]](#)). This data shows that a visual bias helps models ignore this shortcut, whereas an auditory bias makes them more susceptible to it. The detrimental auditory bias is suppressed during the visual only tasks (Fig. 2e [\[10\]](#)), but when forced to switch between both auditory and visual tasks, the auditory bias was more apparent.

Our results show that the architectural biases imparted by feedback are even easier to observe in tasks that require flexible switching, and that the brainlike model and its visual bias could be advantageous over drive-only models in certain scenarios, if for example there is a data imbalance. In contrast, an auditory bias hampered the models for the scenarios studied here.

2.6 Functional specializations of model regions

Different regions of the model are active at different time steps in our simulations, depending on where the region sits in the hierarchy and when feedforward input is received by that region (Fig. 6a [\[11\]](#)). We therefore wanted to understand the impact of top-down feedback on the temporal dynamics of computation in the model across different regions. As such, we next studied the representations of stimuli across time in different regions of the model, in order to gain insight into how the various regions contribute to the tasks studied here. First, to gain a qualitative understanding of representations over time in the models, we projected the hidden states of all

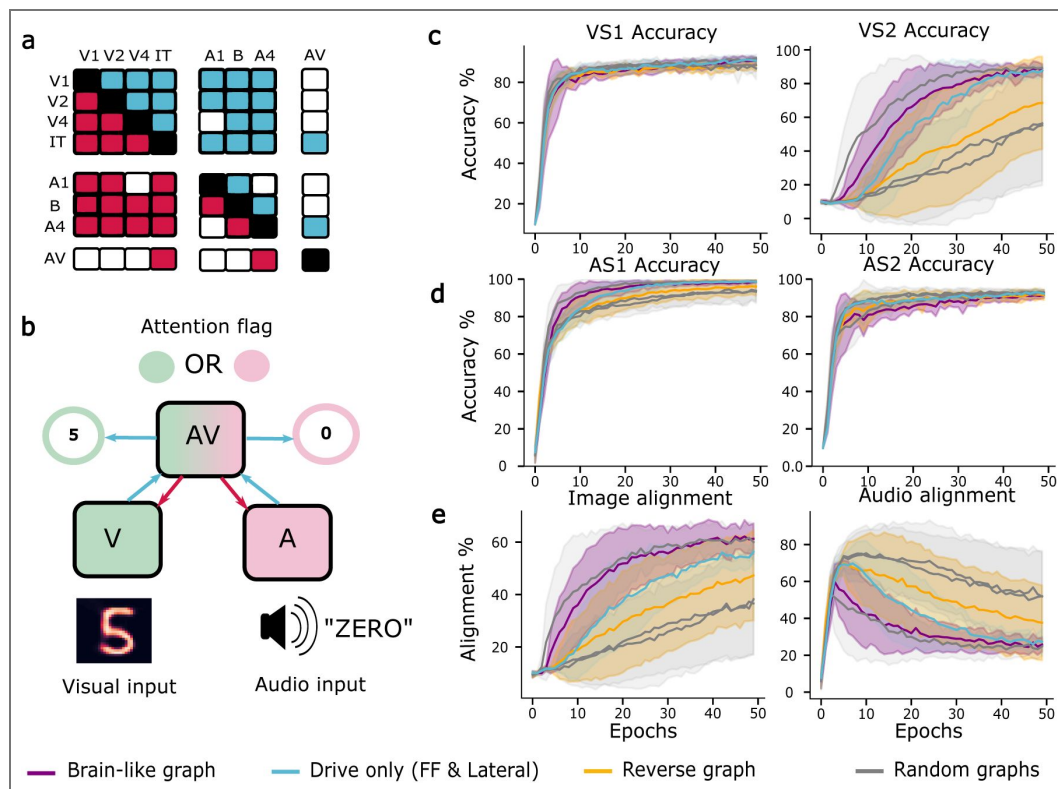


Figure 5. Audiovisual switching task.

a, All models were given a new audiovisual output area (AV) connecting to IT and A4, **b**, The output area receives an attention flag telling which stream of information to attend to (visual or auditory). The models with feedback use with composite feedback. **c-d**, Test performance of models with composite feedback and drive-only models on all tasks. The models were trained simultaneously on all tasks, **e**, Alignment of models given ambiguous visual and ambiguous audio input with differing labels.

regions of the models with composite feedback at different time points onto two dimensions using t-distributed Stochastic Neighbor Embedding (Fig. 6b [↗](#), left). In general, clustered, easily separable representations greatly aid the performance of a neural network in classification tasks like these. We found that different regions showed different clustering of stimuli over time, for example, in the task where the network must classify auditory stimuli and ignore distracting audio stimuli (VS2), the one seed of the brainlike model showed a clear clustering of different stimuli in the IT region, but not in A4, and only moderately in V1 (Fig. 6b [↗](#), left). To quantify the clustering pattern observed across models, we calculated the Neighborhood Hit (NH) score of all datapoints in the latent space at each time step in all regions of the model (Paulovich et al., 2008 [↗](#); Rauber et al., 2017 [↗](#)).

For the visual tasks, in the brainlike and reverse models, we observed distinct, clustered representations in the auditory regions when the auditory stimuli were useful for image identification (Fig. 6c [↗](#), VS1). Conversely, none of the auditory regions in the brainlike and reverse models formed clustered representations in the scenario when the auditory stimuli were irrelevant to the task, outside brief clustering in the A4 in some brainlike models (Fig. 6c [↗](#), VS2). Interestingly, this was not the case with the drive-only model, which exhibited clustering in all auditory regions outside A1 even when the auditory inputs were irrelevant. This demonstrates that in the models with top-down feedback there is a specialization that occurs, such that the auditory regions only clustered the data when necessary, whereas the absence of top-down feedback mechanisms led to more multimodal responses in general.

A similar delineation of function was again evident in the auditory tasks, where the brainlike model had highly clustered representations in V1 and V2 when the visual clue was useful to the task (Fig. 6c [↗](#), AS1), but showed uniformly lower clustering in the visual regions after the first time step if the visual clue had to be ignored (Fig. 6c [↗](#), AS2). Notably, the reverse model did not cluster the visual clue effectively resulting in low clustering in the visual regions during AS1 and a drop in performance (Fig. 3b [↗](#)). In contrast, clustering exhibited by the drive-only model was more variable compared to all models with top-down feedback.

It's important to emphasize that no specific inductive biases were given to any of the regions of the model beyond the inputs they received and whether those inputs were feedforward or feedback. V2 and the auditory belt, for instance, are identical aside from their connectivity in these models. Moreover, all regions were trained end-to-end with the same, unitary loss function. As such, there was no incentive for a non-primary region to develop an exclusive visual or auditory specialization in the absence of the connectivity constraints. An especially illustrative example of this is one of the random models where the V2 region actually developed into an auditory region (Fig. 6c [↗](#), grey line). This model performed consistently well in all tasks. Similarly, the consistently well-performing drive-only model developed drastically different functional specialization between seeds in the auditory tasks (Fig. 6c [↗](#), blue line). Thus, there are multiple different ways to solve these tasks, and different architectures of feedforward and feedback inputs push the models towards these different solutions.

Another illustrative observation was that the brainlike and reverse models developed more predictable specializations in their visual versus auditory regions compared to other models, i.e. visual regions were largely visual and auditory regions were largely auditory (based on their clustering profiles). This is likely because the brainlike and reverse models had visual-to-auditory connections that were either all feedforward or all feedback, respectively. This further confirms that the distinction between feedforward and feedback inputs, as implemented in our models, helps to determine the set of solutions available to the networks and the regional specializations that they develop.

3 Discussion

Neurophysiological and anatomical data suggest that top-down feedback to pyramidal neurons in the neo-cortex has a distinct impact on activity from that of bottom-up feedforward inputs. Specifically, top-down feedback has a more modulatory role, changing the gain of the neurons as

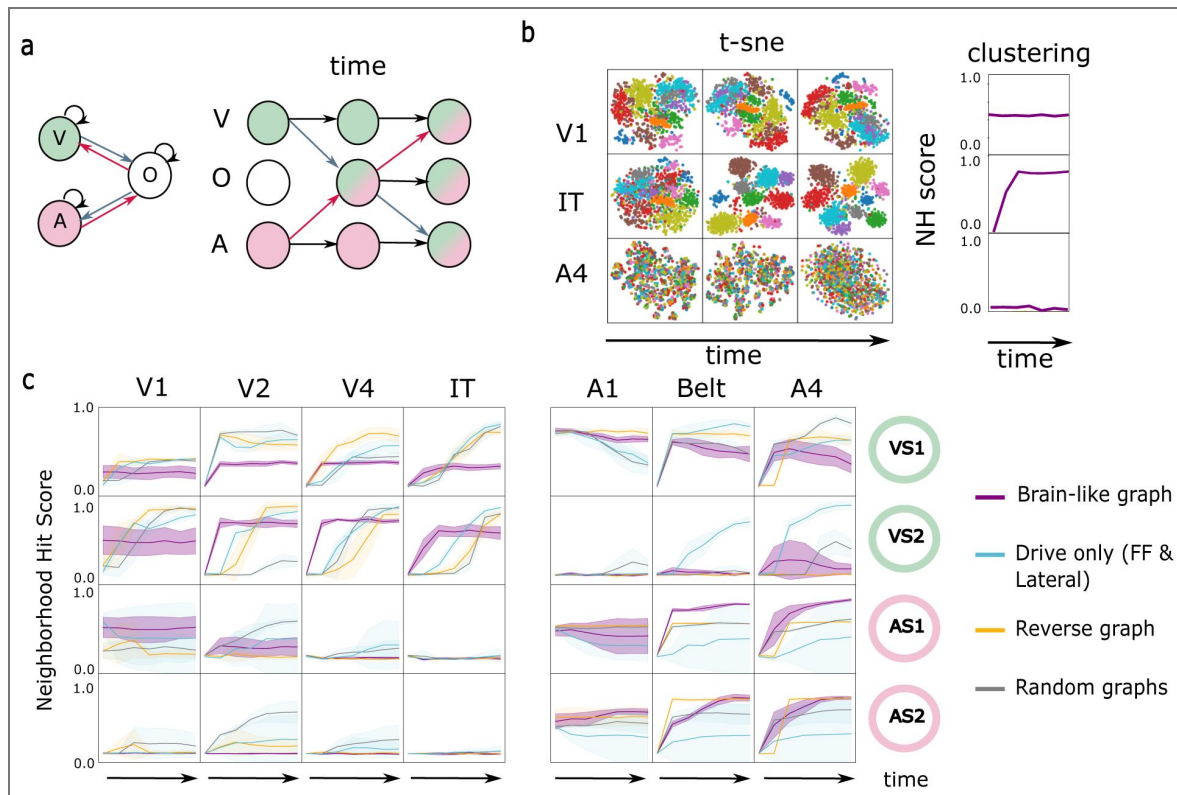


Figure 6. Model activity during multimodal tasks.

a, Information flows through the model from area to area across time. At the first time step, only the primary visual and auditory areas will process information. The areas they feed forward to are activated at the next time step, incorporating top-down information if there is any. **b**, Comparison of t-SNE reduced latent space and clustering metric in three areas of the brainlike model at different time stages on task VS2 (ignore audio stimulus). **c**, Neighborhood Hit scores in all areas of the trained models across time. Trained models were taken from experiments in [Fig 4](#).

well as their spike threshold, rather than directly driving activity (Larkum, 2004 [↗](#); Shai et al., 2015 [↗](#)). In this study, we built what is, to our knowledge, the first hierarchical multi-modal deep neural network architecture based on the brain's anatomy that takes this distinction into account. We then explored how different architectures of feedforward and feedback inputs impacted the behavior of networks on audiovisual integration tasks. We compared a brainlike model, based on human cytoarchitectural data to other models, including a model that reversed the relationship observed in the brain, random models, and a model with only feedforward connections. We found that even in densely connected, identically sized models, different configurations of feedforward and feedback connectivity gave the models different strengths, weaknesses and inductive biases. In particular, deep models with a human brainlike hierarchy exhibited a distinct visual bias, but nevertheless performed well on all audiovisual tasks, qualitatively mimicking a long-known human bias for visual stimuli (Posner et al., 1976 [↗](#); Stokes & Biggs, 2014 [↗](#)). While weakly driving composite feedback improved the performance of some of the lagging models, the visual bias of the brainlike model persisted for both composite and multiplicative feedback. Moreover, all regions of the brainlike model developed their expected functional specializations (i.e. visual regions clustered visual stimuli, and auditory regions clustered auditory stimuli), despite us having given no region-specific bias other than connectivity. This suggests that the profile of feedforward and feedback connectivity of a region helps determine its functional specializations. Altogether, our results demonstrate that the distinction between feedforward and feedback inputs has clear computational implications, and that ANN models of the brain should therefore consider top-down feedback as an important biological feature.

3.1 Computational impact of top-down feedback

Top-down feedback is speculated to play a variety of roles in the cortex. It's known to suppress neural responses to predictable inputs (Nassi et al., 2013 [↗](#); Rao & Ballard, 1999 [↗](#))—an observation that forms the core of the predictive coding framework (Friston and Kiebel, 2009 [↗](#); Mumford, 1992 [↗](#); Rao and Ballard, 1999 [↗](#)). In addition to its predictive function, it's crucial for modulating attention (Debes & Dragoi, 2023 [↗](#)), shaping perception (Manita et al., 2015 [↗](#)), and conveying task-specific context (Li et al., 2004 [↗](#); Liu et al., 2021 [↗](#)). Longrange feedback from the motor (Jordan & Keller, 2020 [↗](#); Leinweber et al., 2017 [↗](#)) and auditory cortex (Garner & Keller, 2021 [↗](#)) carry important contextual information to the early visual cortex.

The many proposed roles of top-down feedback have been explored by various computational models (Choksi et al., 2020 [↗](#); Deco & Rolls, 2004 [↗](#); Huang et al., 2020 [↗](#); Jiang & Rao, 2024 [↗](#); Mittal et al., 2020 [↗](#); Pang et al., 2021 [↗](#); Wen et al., 2018 [↗](#)), but these previous studies have not generally incorporated the multiplicative, modulatory role of top-down feedback. There are a few important, recent exceptions. First, Naumann et al., 2022 [↗](#) examined the impact of a top-down feedback that modulated the feedforward synapses, showing that modulatory feedback can help to recover source signals from a noisy environment. Second, Wybo et al., 2022 [↗](#) used a composite feedback mechanism and biophysical modeling to show that modulatory feedback can help neurons to flexibly solve multiple linearly inseparable problems through Hebbian plasticity. Next, Islah et al., 2023 [↗](#) showed that multiplicative feedback can provide crucial contextual information to a neural network, allowing it to disambiguate challenging stimuli. Most recently, Tsai et al., 2024 [↗](#) showed that modulatory feedback enhances task-relevant sensory signals in a computational model of S1 and IOFC. Our study adds to this previous work by incorporating modulatory top-down feedback into deep, convolutional, recurrent networks that can be matched to real brain anatomy. Importantly, using this framework we could demonstrate that the specific architecture of top-down feedback in a neural network has important computational implications, endowing networks with different inductive biases.

One other potential computational role for top-down feedback is to provide a credit assignment signal (Greedy et al., 2022 [↗](#); Guerguiev et al., 2017 [↗](#); Lee et al., 2015 [↗](#); Payeur et al., 2021 [↗](#); Roelfsema & Ooyen, 2005 [↗](#); Sacramento et al., 2018 [↗](#)). There is some experimental evidence supporting this role for top-down feedback in the brain (Bittner et al., 2017 [↗](#); Doron et al., 2020 [↗](#);

Francioni et al., 2023 [↗](#)), but it is not yet a widely accepted function for top-down inputs. Nonetheless, if additional experimental evidence supports this role for top-down feedback, it will be critical to also incorporate the credit assignment function into future models.

3.2 Testable predictions of audiovisual integration

The idea that auditory cortical regions are higher on the computational hierarchy than visual regions in the human brain is a longstanding, if somewhat controversial hypothesis (King & Nelken, 2009 [↗](#)). Neurons in V1 respond strongly to very low-level features, such as oriented gratings (Hubel & Wiesel, 1962 [↗](#)). Meanwhile, neurons in A1 respond more strongly to complex, modulated, and relatively long stimuli compared to pure tones and frequency sweeps (Wang et al., 2005 [↗](#)). One reason for this may be that the subcortical auditory system in mammals is extensive, and the synaptic distance between the cochlea and A1 is consequently greater than that between the retina and V1. In-line with this, previous modeling work with deep networks showed that the primary auditory area matches intermediate regions of trained deep networks most closely (Kell et al., 2018 [↗](#)) while primary visual regions match early layers (Cadena et al., 2019 [↗](#); Khaligh-Razavi & Kriegeskorte, 2014 [↗](#); Yamins & DiCarlo, 2016 [↗](#)). Our study suggests that if the hypothesis is true, then there are functional consequences that flow from the architecture of the human brain, most notably, a bias towards relying on visual inputs more readily to resolve ambiguities. This is one clear, experimental prediction that our models make.

However, while humans are generally thought to be a visually-dominant species, some individuals with musical training report a behaviorally and functionally distinct auditory bias (Giard & Peronnet, 1999 [↗](#)). The proportion of feedforward and feedback connectivity between the visual and auditory regions may explain these individual differences. This represents another testable prediction flowing from our study, which could be studied in humans by examining the information flow between auditory and visual regions during an audiovisual task (example technique in Pines et al., 2023 [↗](#)). If confirmed, the hypothesis provides a concrete, developmentally flexible mechanism for the emergence of human visual bias.

Moreover, the link between connectivity and functional specialization can be studied further within the same framework to produce testable hypotheses about the effects of cortical lesions, implant responses and stroke recovery. Previous large-scale computational models of lesions lend great insight into their effects on brain function, but cannot readily probe their behavioral effects (Alstott et al., 2009 [↗](#); Martínez-Molina et al., 2024 [↗](#)). In contrast, the ANNs used in our study can be trained on very complex tasks via backpropagation, and thus could provide direct insight into the potential impact of lesions on task performance and behavior. In general, with intelligent selection of architectures guided by anatomy and tasks selected based on real experimental conditions, ANNs with modulatory top-down feedback and brainlike connectivity could be used to generate many predictions about functional connectivity and lesion recovery.

3.3 Limitations and future directions

We made various simplifying assumptions about connectivity in the human brain to build the brain-based model. Feedforward-feedback directionality of a connection is determined in animals using tracer injections, but in human brains, it must be determined using proxy measures. To determine the directionalities in our model, we used externopyramidalisation—the relative thickness and differentiation of supragranular layers, known to be highest at the bottom of the sensory hierarchy. While our measure replicates expected hierarchical ordering of brain regions based on previous literature, visual regions exclusively sending feedforward information to auditory regions in the human brain is an extrapolation of available cytoarchitectural data.

While the visual bias of the brainlike model was evident across all scenarios, the reverse model did not display a similar task-agnostic auditory bias (Fig. 2e [↗](#)). This is likely because the auditory dataset in our study is smaller, less variable and thus easier to learn than the visual dataset, causing certain models to shortcut their training and rely excessively on it (Fig. 3b [↗](#) and Fig. 3e [↗](#)). A visual bias helps models ignore this shortcut, whereas an auditory bias makes them more susceptible to it and hampers a model's ability to ignore audio input on visual tasks (Fig. 2d [↗](#), Fig.

5b [↗](#)). This points to another limitation in our study, namely the use of relatively simple stimuli and small datasets. Ideally, future work would use rich, natural audiovisual inputs and large datasets to train the networks. This may lead to some different behaviors in the models. Moreover, our results relate to other findings that task demands often supersede architectural choices in computational modeling (Lindsay et al., 2022 [↗](#)). Nonetheless, despite these limitations, our model shows clearly that within certain tasks the biases imparted by architectural choices have important implications that interact with the data.

Another important limitation to note is that many aspects of our model are not biologically plausible. Whether a connection is feedback or feedforward is fixed at the start of the experiments, whereas connectivity is more flexible and task-based in biological brains. Moreover, our model uses end-to-end backpropagation and supervised learning. However, our models are not only applicable in these cases - one could easily train the ANNs we designed with different learning algorithms. As well, the models we developed are agnostic to training method and can easily be trained with self-supervised and reinforcement learning tasks. But, we believe that the ability to train our models with backpropagation is important, because that will allow them to learn a wide range of complicated, more natural tasks in future research. We have released the codebase to construct these models to facilitate such research.

Finally, another key limitation in this study is that we did not compare our models directly to human neural data. Our results show clearly that the models' internal representations are altered by top-down feedback (Fig. 6 [↗](#)), so we would expect it to also have an impact on the ability of the models to match the representations in real brains. But, we leave this as future work, which is made easier by the release of the codebase.

In summary, our study shows that modulatory top-down feedback and the architectural diversity enabled by it can have important functional implications for computational models of the brain. We believe that future work examining brain function with deep neural networks should therefore consider incorporating top-down modulatory feedback into model architectures when appropriate. More broadly, our work supports the conclusion that both the cellular neurophysiology and structure of feedback inputs have critical functional implications that need to be considered by computational models of brain function.

4 Methods

4.1 Model

Each area/layer of the model is a modified Convolutional Gated Recurrent Unit (ConvGRU) cell. A Gated Recurrent Unit is a standard neural network with locally recurrent connections. The processes are identical to that of a typical GRU with the exception of the linear operators, which are replaced by convolutions (represented by the star) as per Ballas et al., 2016 [↗](#), and the topdown signal (m), which modulates the hidden state after it has been reset and combined with the bottom-up input.

The tasks in this paper are non-sequential; the image and the audio stimuli are presented at the same time to V1 and A1 respectively. At each time step, the areas receiving a bottom-up feedforward input (h_{l-1}) get activated. The feedforward input is combined with the hidden state memory of the area (h_l), and modulated by the excitatory topdown signal (m), which is derived from the feedback input (h_{l+1}) if it exists. In case there is no feedback input (i.e the higher order areas have not been active yet), m is a matrix of ones. Resets and updates (Equations 1 [↗](#) and 2 [↗](#)) are features of the GRU that implement local recurrence. The full equations are presented below.

$$r_l = \sigma(W_r^* \odot [h_{l-1}; h_l] + b_r) \quad (1)$$

$$u_l = \sigma(W_u^* \odot [h_{l-1}; h_l] + b_u) \quad (2)$$

$$m = \text{ReLU}(W_m(h_{l+1}) + b_m) + 1 \quad (3)$$

$$c = (W_c^* \odot [h_{l-1}; r_l \odot h_l] + b_c) \odot m \quad (4)$$

$$h_l = (1 - u_l) \odot h_l + u_l \times \tanh(c) \quad (5)$$

where r_l is the reset gate for layer l , u_l is the update gate for layer l , h_l is the hidden state of layer l , h_{l-1} is the feedforward input from the previous layer, h_{l+1} is the feedback input from the next layer, m is the modulatory top-down signal, c is the candidate activation, $[x; y]$ denotes concatenation along the channel dimension, $\star$ denotes convolution, $\odot$ denotes element-wise multiplication, and W and b represent learnable weights and biases respectively.

The input size of the regions decreased as they moved up the sensory-association hierarchy, i.e. V1 and A1 received 32x32 input, while V2 and the auditory belt received 16x16 input. All regions of the model had a hidden state channel size of 10, matching the number of classes. The small size of the model is to make differences in performance clearer for easier tasks, but as shown in Fig. S2 [S2](#), larger models (with hidden state size of 32) have similar image/auditory biases with a higher performance ceiling.

All feedforward inputs to an area were combined and projected into the correct shape by a convolutional projection layer. Feedback inputs were similarly reshaped by a convolutional projection layer to a shape that matches the hidden state of the region.

All models presented in the main figures had the same number of parameters within the two significant figures range (280K), since the number of connections, number of nodes and sizes of hidden states are the same for all of them with only directionality of connections changing. Small differences (1K) in size exist between multiplicative and composite feedback models due the extra driving input from high-order areas that are characteristic of the composite models.

4.1.1 Composite feedback

When using composite feedback, the top-down signal was split into the modulatory signal m_{mod} matching the shape of the feedforward input, and the driving signal m_d , which has ten times fewer channels than the feedforward input. In other words, in addition to multiplicative modulation, the feedback signal provided a driving signal at one tenth the strength of the feedforward driving signal. Equation 4 [Equation 4](#) was then adjusted to incorporate the two different types of top-down signals:

$$c = \tanh((W_c^* n \times [h_{l-1}; r_l \times h_l; m_d] + b_c) \times m_{mod}) \quad (6)$$

4.1.2 Classification

For supervised classification tasks, the user must designate an output region (IT in visual tasks and A4 in auditory tasks). The hidden states of the output region are fed to two-layer multilayer perceptron (MLP) trained at the same time as the model, which then outputs a classification.

4.2 Bases of the human brainlike model

We based the brainlike model on human structural connectivity and histological data. For all human data, regions of interest were determined using the multimodal HCP parcellation (Glasser et al., 2016 [S2](#)). Four classic regions in the ventral visual system was chosen to represent the visual cortex: V1, V2, V4, and PIT. Three auditory areas were chosen to represent the auditory cortex: A1, parabelt (part of the belt bordering A4), and A4, a portion of Brodmann area 22 that activated by language tasks and is thought to correspond to area Te3 (Morosan et al., 2005 [S2](#)).

To determine the overall connectivity between regions, we used the diffusion tractography data of 50 adult subjects from the Microstructure-Informed Connectomics (MICA-MICs) dataset (Royer et al., 2022 [S2](#)). We averaged the connectivity between all pair of regions across subjects and generated a group-average binary connectivity matrix using distance-based thresholding (Betzel et al., 2019 [S2](#)).

To determine the directionality of connectivity between regions, we used the BigBrain quantitative 3D laminar atlas of the human cerebral cortex (Amunts et al., 2013 [S2](#); Wagstyl et al., 2020 [S2](#)). We used externopyramidalisation (Sanides, 1962 [S2](#)), the relative size of supragranular neurons compared to infragranular neurons, as the proxy for the hierarchical position of area. In mammals including humans, it is high in regions that send more supragranular feedforward projections and lower in areas that send more infragranular feedback projections (Goulas et al., 2018 [S2](#)). This

Parameter	Definition
r_l	Reset gate for layer l
u_l	Update gate for layer l
h_l	Hidden state of layer l
h_{l-1}	Feedforward input from the previous layer
h_{l+1}	Feedback input from the next layer
m	Modulatory top-down signal
m_{mod}	Modulatory component of top-down signal (composite feedback)
m_d	Driving component of top-down signal (composite feedback)
c	Candidate activation
W_r	Learnable weight matrix for reset gate
W_u	Learnable weight matrix for update gate
W_c	Learnable weight matrix for candidate activation
W_m	Learnable weight matrix for modulatory signal
b_r	Bias vector for reset gate
b_u	Bias vector for update gate
b_c	Bias vector for candidate activation
b_m	Bias vector for modulatory signal
σ	Sigmoid activation function
$\tanh$	Hyperbolic tangent activation function
ReLU	Rectified Linear Unit activation function
$[x; y]$	Concatenation along the channel dimension
$\star$	Convolution operation
$\odot$	Element-wise multiplication

Table 1. Parameters for the modified ConvGRU equations

value can be estimated from histological data using the relative laminar thickness and staining intensity of supragranular layers, a process described by Paquola et al., 2020 [\[14\]](#). Similar to the method described in that paper, we rescaled supragranular staining intensity (in) and thickness values to a range of 0 to 1, and used the normalized peak intensity from 20 sample locations (n) and relative thickness of supragranular layers to calculate externopyramidisation.

$$\text{externopyramidisation} = \frac{\max(in)}{\frac{1}{n} \sum_{i=1}^n in_i} \times \frac{1 - \text{thickness}_{supra}}{\text{thickness}_{total}} \quad (7)$$

Regions with higher externopyramidisation values were configured to send feedforward connections to and receive feedback from regions with lower externopyramidisation values.

4.3 Stimuli generation and preprocessing

We used MNIST handwritten digit database as the unambiguous visual stimuli and the Free Spoken Digit Dataset (FSDD) as the unambiguous auditory stimuli. The visual stimuli were minimally preprocessed, while the audio was Fourier-transformed and converted to log scale to generate uniformly sized mel-spectrograms. The ambiguous visual stimuli were created by Islah et al., 2023 [\[15\]](#) and used with permission. Details of it can be found in the cited paper. The ambiguous visual stimuli all have two equally possible labels, capping the performance of classifiers at around 50 percent if they are not given additional clues.

4.3.1 Ambiguous auditory stimuli

The ambiguous auditory stimuli were generated in a similar manner to the ambiguous visual stimuli. We trained a conditional variational autoencoder (CVAE) (Sohn et al., 2015 [\[16\]](#)) to project mel-spectrograms of FSDD data into a 32 dimensional latent space. We then sampled the Euclidean mean of two random data in the latent space with differing labels, and used the decoder of the trained CVAE to deconvolve it into a mel-spectrogram. The resulting ambiguous digit often retained the features of one digit, but lost all features of the other it was “mixed” with. As such, while the ambiguous visual stimuli have two equally possible interpretations by design, the ambiguous auditory stimuli have a single possible interpretation that is nevertheless difficult to identify without additional clues. To ensure a balance of recognizability and ambiguity in the auditory data, we trained a separate softmax classifier on mel spectrograms of unambiguous FSDD data. The softmax classifier output a prediction between 0 and 1.0 (full certainty) for each label. We asked the classifier to predict the labels of each newly generated ambiguous mel spectrogram, and kept only the data and labels that caused the classifier to output a prediction between 0.45 and 0.55. We reiterated the process until we had 4000 unique ambiguous auditory stimuli, the size of the holdout portion of original FSDD dataset. Due to the higher power of the ConvGRU-based models used in our experiments compared to the softmax classifier used for quality control, the models used in the experiments are able to decipher the ambiguous audio up to 80 percent accuracy without clues.

4.4 Audiovisual task training

We combined the four types of stimuli into fixed audiovisual datasets, where each datapoint consisted of a pair of audio and visual stimuli and their corresponding labels. We held out 15 percent of each dataset for testing before creating the audiovisual datasets, meaning no image or audio during testing was encountered during training. We calculated the gradient in minibatches of 32, and used the Adam optimizer at learning rate 0.0001 to train the model. All models were trained for 50 epochs. For each experimental condition, we ran 10 different seeds. We additionally tested learning rates of 0.001, 0.01 and 0.1 after getting the primary results. The models show the same behaviors and preferences under all rates except the highest learning rate of 0.1, where all models train poorly.

4.4.1 Training tasks

For tasks VS1 and VS2, we trained the models on a shuffled mix of datasets UAM (unambiguous audio, ambiguous image, matching label) and UUN (unambiguous audio, unambiguous image, mismatched label). We set the label of the image as the target and calculated the cross-entropy loss between it and the model prediction.

For tasks AS1 and AS2, we trained the models on a shuffled mix of datasets AUM (unambiguous audio-ambiguous image, matching label) and UUN (unambiguous audio, unambiguous image, mismatched label). We set the audio labels as the target and otherwise trained the models under the same parameters as the visual tasks.

4.4.2 Control tasks and alignment

Datasets AUN (ambiguous audio, unambiguous image, nonmatching label) and UAN (unambiguous audio, ambiguous image, nonmatching label) were used only for testing in scenarios AS3 and VS3 respectively. The visual alignment of a model is determined as the percentage of time when the model outputs the label of the visual stimuli. Same is true for the auditory dataset. The test-only datasets were also generated solely from holdout stimuli.

In addition, models were separately trained and evaluated on unambiguous MNIST and FSDD data to establish their baseline performance (scenarios VS4 and AS4).

4.4.3 Flexible task training

For the flexible task-switching experiments, the models are given a new output region and trained on datasets AUM, UAM, and UUN. Its output region is given a binary attention flag indicating the input stream to attend to.

After training, the image and auditory alignment of the models were tested using the previously unseen AAN (ambiguous audio, ambiguous image, nonmatching label) dataset.

4.4.4 Process time

All models have a unique hyperparameter called process time, which roughly determines how many computational steps the model can take before it must output an answer. A process time of 1, for instance, will mean that only the primary areas A1 and V1 will have received any feedforward input before the model generates an output. For all audiovisual tasks, we picked a process time equal to the number of areas in the model (7).

4.5 Regional activity analysis

We fetched the hidden state representation of all regions of the trained models at 7 different process times from 1-7 while they performed the tasks they were trained on. We then measured the quality of clustering in the latent space of each model region and time point by calculating the Neighborhood Hit (NH) score (Paulovich et al., 2008 [↗](#); Rauber et al., 2017 [↗](#)). The NH score is the mean percentage of neighbors r_j for each datapoint n who share the same label $C[x_i]$. It's calculated as below, using $k=4$ for every task.

$$NH(k) = \frac{1}{n} \sum_{i=1}^n \left(\frac{1}{k} \sum_{j=1}^k l_j \right), \quad l_j = \begin{cases} 1 & \text{if } r_j \in C[x_i] \\ 0 & \text{else} \end{cases} \quad (8)$$

4.6 Statistical analysis

Mean final accuracies for multiplicative and composite feedback were compared using Welch's t-test. Error bars and bands represent standard error of the mean, unless stated otherwise. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$; n.s., not significant.

4.7 Code availability

The toolbox used to convert graphs to top-down recurrent neural networks (Connectome-to-Model) is publicly available at <https://github.com/masht18/connectome-to-model>. The task training scripts and graphs are also available at the same repository.

Supplemental

	Brainlike	Reverse	Rand1	Rand2	Rand3
VS1	0.868 ± 0.009	0.914 ± 0.011	0.885 ± 0.013	0.880 ± 0.030	0.836 ± 0.011
VS2	0.933 ± 0.006	0.945 ± 0.013	0.909 ± 0.021	0.884 ± 0.058	0.800 ± 0.052
VS3 (image align)	0.446 ± 0.007	0.440 ± 0.006	0.436 ± 0.010	0.424 ± 0.016	0.406 ± 0.027
VS3 (audio align)	0.216 ± 0.008	0.242 ± 0.011	0.237 ± 0.007	0.247 ± 0.008	0.255 ± 0.029
VS4 (image recognition)	0.964 ± 0.003	0.939 ± 0.007	0.967 ± 0.002	0.907 ± 0.025	0.959 ± 0.009

Table 2. Mean and standard deviation of final epoch accuracy across VS tasks for small models with multiplicative feedback (Fig. 2).

Table 3. Mean and standard deviation of final epoch accuracy across AS tasks for small models with multiplicative feedback (Fig. 3)

	Brainlike	Reverse	Rand1	Rand2	Rand3
AS1	0.972 ± 0.006	0.880 ± 0.046	0.968 ± 0.039	0.879 ± 0.047	0.887 ± 0.054
AS2	0.944 ± 0.016	0.937 ± 0.009	0.945 ± 0.005	0.945 ± 0.015	0.946 ± 0.007
AS3 (image alignment)	0.509 ± 0.066	0.164 ± 0.102	0.452 ± 0.181	0.165 ± 0.095	0.204 ± 0.137
AS3 (audio alignment)	0.326 ± 0.053	0.775 ± 0.155	0.428 ± 0.207	0.764 ± 0.157	0.729 ± 0.187
AS4 (audio recognition)	0.928 ± 0.012	0.938 ± 0.010	0.936 ± 0.016	0.919 ± 0.022	0.922 ± 0.015

Table 4. Mean and standard deviation across VS tasks for small models with composite feedback (Fig. 4).

	Brainlike	Reverse	Rand1	Rand2	Rand3
VS1	0.928 ± 0.014	0.907 ± 0.007	0.946 ± 0.007	0.945 ± 0.008	0.941 ± 0.012
VS2	0.949 ± 0.010	0.941 ± 0.012	0.962 ± 0.007	0.671 ± 0.402	0.811 ± 0.317
VS3 (image align)	0.442 ± 0.007	0.445 ± 0.012	0.450 ± 0.006	0.329 ± 0.163	0.384 ± 0.128
VS3 (audio align)	0.247 ± 0.012	0.235 ± 0.011	0.247 ± 0.008	0.483 ± 0.326	0.368 ± 0.260
AS1	0.967 ± 0.007	0.897 ± 0.054	0.964 ± 0.038	0.892 ± 0.053	0.877 ± 0.059
AS2	0.928 ± 0.013	0.928 ± 0.008	0.940 ± 0.011	0.935 ± 0.010	0.933 ± 0.014
AS3 (image align)	0.453 ± 0.073	0.200 ± 0.112	0.444 ± 0.137	0.193 ± 0.128	0.192 ± 0.130
AS3 (audio align)	0.368 ± 0.065	0.722 ± 0.167	0.423 ± 0.164	0.741 ± 0.173	0.741 ± 0.189

Figure S1. Connectivity of the three random models

The connectivity of the random models were consistent throughout all experiments

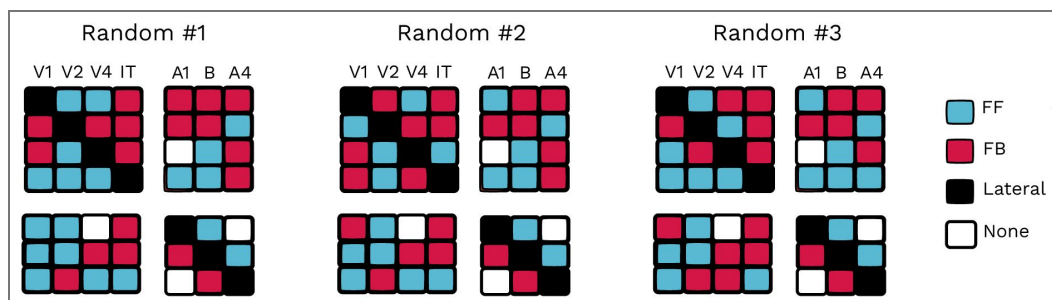


Figure S2. Performance of larger model (hidden state size 32) on AS tasks.

Performance is increased across the board, but difference in stimuli preference is still visible

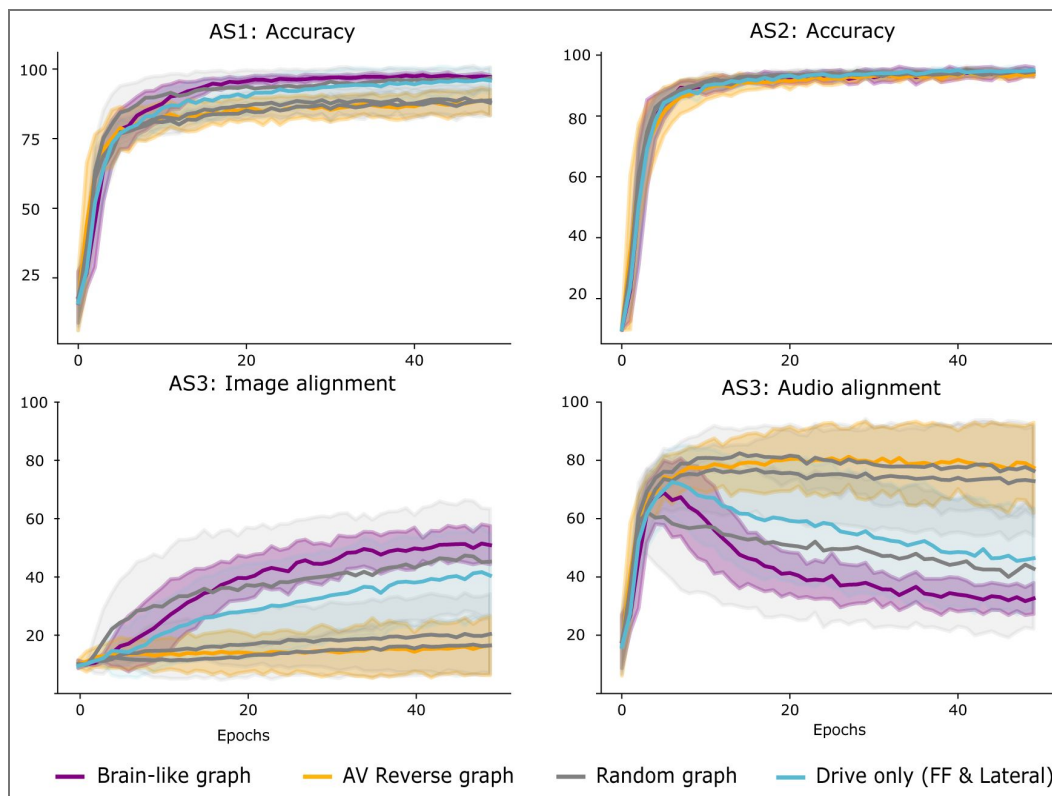


Figure S3. Effect of process time on AS task performance

Final epoch accuracy in model given short (3P), standard (7P, equal to number of model layers) and long (10P) processing times when given **a** ambiguous audio, helpful visual clue, **b** clean audio and distracting image, **c,d** ambiguous audio and nonmatching visual input

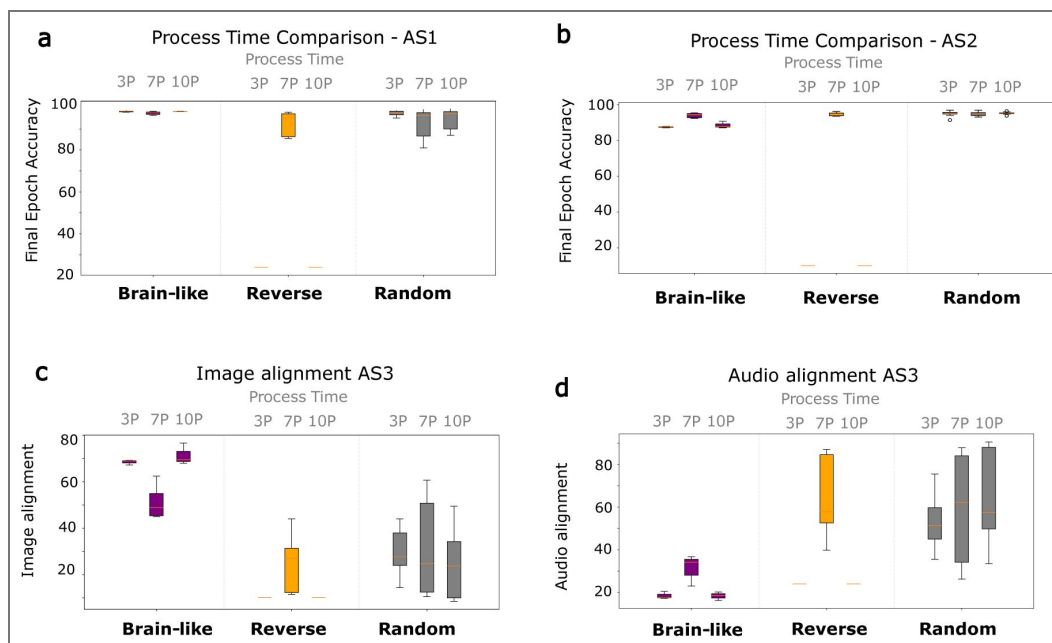


Figure S4. Models with multiplicative feedback on audiovisual switching task.

The task and the drive-only (blue) models are the same between this figure and Figure 5.

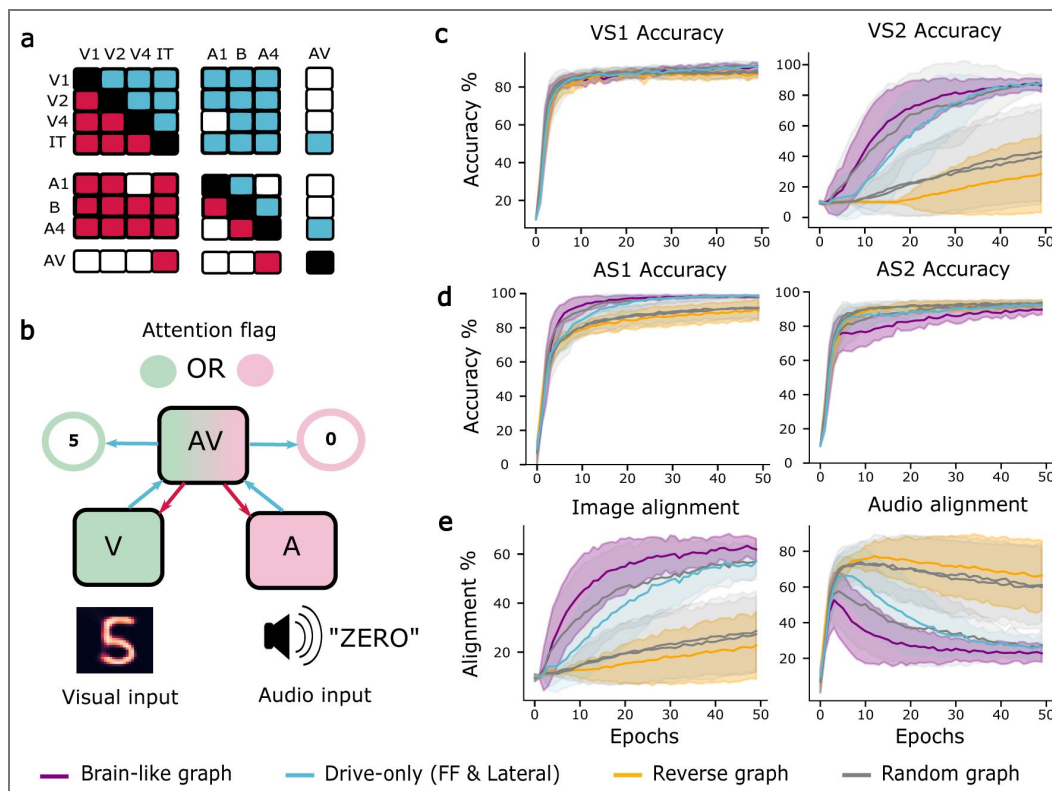


Table 5. Mean and standard deviation of final epoch accuracy across audiovisual switching tasks (Fig. 5).

	Brainlike	Reverse	Rand1	Rand2	Rand3
VS1	0.909 ± 0.020	0.869 ± 0.023	0.912 ± 0.022	0.883 ± 0.027	0.875 ± 0.028
VS2	0.864 ± 0.042	0.285 ± 0.254	0.874 ± 0.055	0.399 ± 0.306	0.430 ± 0.319
AS1	0.982 ± 0.008	0.900 ± 0.058	0.983 ± 0.014	0.917 ± 0.062	0.913 ± 0.065
AS2	0.897 ± 0.029	0.925 ± 0.023	0.922 ± 0.018	0.937 ± 0.017	0.937 ± 0.019
Image align	0.619 ± 0.044	0.228 ± 0.138	0.570 ± 0.073	0.273 ± 0.155	0.286 ± 0.163
Audio align	0.228 ± 0.050	0.665 ± 0.193	0.273 ± 0.083	0.611 ± 0.206	0.600 ± 0.242

Table 6. Mean and standard deviation of final epoch accuracy for big models on AS tasks (Fig. S2).

	Brainlike	Reverse	Rand1	Rand2	Rand3
AS1	0.987 ± 0.004	0.949 ± 0.036	0.984 ± 0.008	0.979 ± 0.008	0.977 ± 0.010
AS2	0.952 ± 0.011	0.949 ± 0.009	0.953 ± 0.011	0.962 ± 0.012	0.971 ± 0.002
AS3 (image align)	0.556 ± 0.036	0.201 ± 0.128	0.358 ± 0.102	0.366 ± 0.045	0.353 ± 0.105
AS3 (audio align)	0.280 ± 0.015	0.689 ± 0.203	0.477 ± 0.124	0.431 ± 0.045	0.484 ± 0.135

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Peer reviews

Reviewer #1 (Public review):

Summary:

Here, the authors aim to investigate the potential improvements of ANNs when used to explain brain data using top-down feedback connections found in the neocortex. To do so, they use a retinotopic and tonotopic organization to model each subregion of the ventral visual (V1, V2, V4, and IT) and ventral auditory (A1, Belt, A4) regions using Convolutional Gated Recurrent Units. The top-down feedback connections are inspired by the apical tree of pyramidal neurons, modeled either with a multiplicative effect (change of gain of the activation function) or a composite effect (change of gain and threshold of the activation function).

To assess the functional impact of the top-down connections, the authors compare three architectures: a brain-like architecture derived directly from brain data analysis, a reversed architecture where all feedforward connections become feedback connections and vice versa, and a random connectivity architecture. More specifically, in the brain-like model the visual regions provide feedforward input to all auditory areas, whereas auditory areas provide feedback to visual regions.

First, the authors found that top-down feedback influences audiovisual processing and that the brain-like model exhibits a visual bias in multimodal visual and auditory tasks. Second, they discovered that in the brain-like model, the composite integration of top-down feedback, similar to that found in the neocortex, leads to an inductive bias toward visual stimuli, which is not observed in the feedforward-only model. Furthermore, the authors found that the brain-like model learns to utilize relevant stimuli more quickly while ignoring distractors. Finally, by analyzing the activations of all hidden layers (brain regions), they found that the feedforward and feedback connectivity of a region could determine its functional specializations during the given tasks.

Strengths:

The study introduces a novel methodology for designing connectivity between regions in deep learning models. The authors also employ several tasks based on audiovisual stimuli to support their conclusions. Additionally, the model utilizes backpropagation of error as a learning algorithm, making it applicable across a range of tasks, from various supervised learning scenarios to reinforcement learning agents. Conversely, the presented framework offers a valuable tool for studying top-down feedback connections in cortical models. Thus, it is a very nice study that can also give inspiration to other fields (machine learning) to start exploring new architectures.

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Reviewer #2 (Public review):

Summary:

This work addresses the question whether artificial deep neural network models of the brain could be improved by incorporating top-down feedback, inspired by the architecture of neocortex.

In line with known biological features of cortical top-down feedback, the authors model such feedback connections with both, a typical driving effect and a purely modulatory effect on the activation of units in the network.

To assess the functional impact of these top-down connections, they compare different architectures of feedforward and feedback connections in a model that mimics the ventral visual and auditory pathways in cortex on an audiovisual integration task.

Notably, one architecture is inspired by human anatomical data, where higher visual and auditory layers possess modulatory top-down connections to all lower-level layers of the same modality, and visual areas provide feedforward input to auditory layers, whereas auditory areas provide modulatory feedback to visual areas.

First, the authors find that this brain-like architecture imparts the models with a light visual bias similar to what is seen in human data, which is the opposite in a reversed architecture, where auditory areas provide feedforward drive to the visual areas.

Second, they find that, in their model, modulatory feedback should be complemented by a driving component to enable effective audiovisual integration, similar to what is observed in neural data.

Overall, the study shows some possible functional implications when adding feedback connections in a deep artificial neural network that mimic some functional aspects of visual perception in humans.

Strengths:

The study contains innovative ideas, such as incorporating an anatomically inspired architecture into a deep ANN, and comparing its impact on a relevant task to alternative architectures.

Moreover, the simplicity of the model allows it to draw conclusions on how features of the architecture and functional aspects of the top-down feedback affects performance of the network.

This could be a helpful resource for future studies of the impact of top-down connections in deep artificial neural network models of neocortex.

Weaknesses:

Some claims not yet supported.

The problem is that results are phrased quite generally in the abstract and discussion, while the actual results shown in the paper are very specific to certain implementations of top-down feedback and architectures. This could lead to misunderstanding and requires some revisions of the claims in the abstract and discussion (see below).

"Altogether our findings demonstrate that modulatory top-down feedback is a computationally relevant feature of biological brain..."

This claim is not supported, since no performance increase is demonstrated for modulatory feedback. So far, only the second half of the sentence is supported: "...and that incorporating it into ANNs affects their behavior and constrains the solutions it's likely to discover."

"This bias does not impair performance on the audiovisual tasks."

This is only true for the composite top-down feedback that combines driving and modulatory effects, whereas modulatory feedback alone can impair the performance (e.g., in the visual tasks VS1 and VS2). The fact that modulatory feedback alone is insufficient in ANNs to enable effective cross-modal integration and requires some driving component is actually very interesting, but it is not stressed enough in the abstract. This is hinted at in the following sentence, but should be made more explicitly:

"The results further suggest that different configurations of top-down feedback make otherwise identically connected models functionally distinct from each other, and from traditional feedforward and laterally recurrent models."

"Here we develop a deep neural network model that captures the core functional properties of top-down feedback in the neocortex" -> this is too strong, take out "the", because very likely there are other important properties that are not yet incorporated.

"Altogether, our results demonstrate that the distinction between feedforward and feedback inputs has clear computational implications, and that ANN models of the brain should therefore consider top-down feedback as an important biological feature."

This claim is still not substantiated by evidence provided in the paper. First, the wording is a bit imprecise, because mechanistically, it is not really the feedforward versus feedback (a purely feedforward model is not considered at all in the paper), but modulatory versus driving. Moreover, the second part of the sentence is problematic: The results imply that, computationally/functionally, driving connections are doing the job, while modulatory feedback does not really seem to improve performance (best case, it does not do any harm). It is true that it is a feature that is inspired by biology, but I don't see why the results imply that (modulatory) top-down feedback should be considered in ANN models of the brain. This would require to show that such models either improve performance, or do improve the ability to fit neural data, both which are beyond the scope of the paper.

The same argument holds for the following sentence, which is not supported by the results of the paper:

"More broadly, our work supports the conclusion that both the cellular neurophysiology and structure of feed-back inputs have critical functional implications that need to be considered by computational models of brain function."

Additional supplementary material required

Although the second version checked the influence of processing time, this was not done for the most important figure of the paper, Figure 4. A central claim in the abstract "This bias does not impair performance on the audiovisual tasks" relies on this figure, because only with composite feedback the performance is comparable between the "drive-only" and "brain-like" models. Thus, the supplementary Figure 3 should also include the composite networks and drive only network to check the robustness of the claim with respect to process time. This robustness analysis should then also be mentioned in the text. For example, it should be mentioned whether results in these networks are robust or not with respect to process time, whether there are differences between network architectures or types of feedback in general etc.

Moreover, the current analysis for networks with modulatory feedback is a bit confusing. Why is the performance so low for the reverse model for a process time of 3 and 10? This is a very strong effect that warrants explanation. More details should be added in the caption as well. For example, are the models separately trained for the output after 3 and 10 processing steps for the comparison, or just evaluated at these times? Not training these networks separately might explain the low performance for some networks, so ideally networks are trained for each choice of processing steps.

<https://doi.org/10.7554/eLife.105953.2.sa2>

Reviewer #3 (Public review):

Summary:

This study investigates the computational role of top-down feedback in artificial neural networks (ANNs), a feature that is prevalent in the brain but largely absent in standard ANN architectures. The authors construct hierarchical recurrent ANN models that incorporate key properties of top-down feedback in the neocortex. Using these models in an audiovisual integration task, they find that hierarchical structures introduce a mild visual bias, akin to that observed in human perception, not always compromising task performance.

Strengths:

The study investigates a relevant and current topic of considering top-down feedback in deep neural networks. In designing their brain-like model, they use neurophysiological data, such

as externopyramidisation and hierarchical connectivity. Their brain-like model exhibits a visual bias that qualitatively matches human perception.

Weaknesses:

While the model is brain-inspired, it has limited bioplausibility. The model assumes a simplified and fixed hierarchy. The authors acknowledge this limitation in the discussion.

While the brain-like model showed an advantage in ignoring distracting auditory inputs, it struggled when visual information had to be ignored. This suggests that its rigid bias toward visual processing could make it less adaptive in tasks requiring flexible multimodal integration. It hence does not necessarily constitute an improvement over existing ANNs. The study does not evaluate whether the top-down feedback architecture scales well to more complex problems or larger datasets. A valuable future contribution would be to evaluate how the network's behaviour fits to human data.

<https://doi.org/10.7554/eLife.105953.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

Here, the authors aim to investigate the potential improvements of ANNs when used to explain brain data using top-down feedback connections found in the neocortex. To do so, they use a retinotopic and tonotopic organization to model each subregion of the ventral visual (V1, V2, V4, and IT) and ventral auditory (A1, Belt, A4) regions using Convolutional Gated Recurrent Units. The top-down feedback connections are inspired by the apical tree of pyramidal neurons, modeled either with a multiplicative effect (change of gain of the activation function) or a composite effect (change of gain and threshold of the activation function).

To assess the functional impact of the top-down connections, the authors compare three architectures: a brain-like architecture derived directly from brain data analysis, a reversed architecture where all feedforward connections become feedback connections and vice versa, and a random connectivity architecture. More specifically, in the brain-like model the visual regions provide feedforward input to all auditory areas, whereas auditory areas provide feedback to visual regions.

First, the authors found that top-down feedback influences audiovisual processing and that the brain-like model exhibits a visual bias in multimodal visual and auditory tasks. Second, they discovered that in the brain-like model, the composite integration of top-down feedback, similar to that found in the neocortex, leads to an inductive bias toward visual stimuli, which is not observed in the feedforward-only model. Furthermore, the authors found that the brain-like model learns to utilize relevant stimuli more quickly while ignoring distractors. Finally, by analyzing the activations of all hidden layers (brain regions), they found that the feedforward and feedback connectivity of a region could determine its functional specializations during the given tasks.

Strengths:

The study introduces a novel methodology for designing connectivity between regions in deep learning models. The authors also employ several tasks based on audiovisual

stimuli to support their conclusions. Additionally, the model utilizes backpropagation of error as a learning algorithm, making it applicable across a range of tasks, from various supervised learning scenarios to reinforcement learning agents. Conversely, the presented framework offers a valuable tool for studying top-down feedback connections in cortical models. Thus, it is a very nice study that also can give inspiration to other fields (machine learning) to start exploring new architectures.

We thank the reviewer for their accurate summary of our work and their kind assessment of its strengths.

Weaknesses:

Although the study explores some novel ideas on how to study the feedback connections of the neocortex, the data presented here are not complete in order to propose a concrete theory of the role of top-down feedback inputs in such models of the brain.

(1) The gap in the literature that the paper tries to fill in the ability of DL algorithms to predict behavior: "However, there are still significant gaps in most deep neural networks' ability to predict behavior, particularly when presented with ambiguous, challenging stimuli." and "[...] to accurately model the brain."

It is unclear to me how the presented work addresses this gap, as the only facts provided are derived from a simple categorization task that could also be solved by the feedforward-only model (see Figures 4 and 5). In my opinion, this statement is somewhat far-fetched, and there is insufficient data throughout the manuscript to support this claim.

We can see now that the way the introduction was initially written led to some confusion about our goal in this study. Our goal here was not to demonstrate that top-down feedback can enable superior matches to human behaviour. Rather, our goal was to determine if top-down feedback had any real implications for processing ambiguous stimuli. The sentence that the reviewer has highlighted was intended as an explanation for why top-down feedback, and its impact on ambiguous stimuli, might be something one would want to examine for deep neural networks. But, here, we simply wanted to (1) provide an overview of the code base we have created, (2) demonstrate that top-down feedback does impact the processing of ambiguous stimuli.

We agree with the reviewer that if our goal was to improve our ability to predict behaviour, then there was a big gap in the evidence we provided here. But, this was not our goal, and we believe that the data we provide here does convincingly show that top-down feedback has an impact on processing of ambiguous stimuli. We have updated the text in the introduction to make our goals more clear for the reader and avoid this misunderstanding of what we were trying to accomplish here. Specifically, the end of the introduction is changed to:

"To study the effect of top-down feedback on such tasks, we built a freely available code base for creating deep neural networks with an algorithmic approximation of top-down feedback. Specifically, top-down feedback was designed to modulate ongoing activity in recurrent, convolutional neural networks. We explored different architectural configurations of connectivity, including a configuration based on the human brain, where all visual areas send feedforward inputs to, and receive top-down feedback from, the auditory areas. The human brain-based model performed well on all audiovisual tasks, but displayed a unique and persistent visual bias compared to models with only driving connectivity and models with different hierarchies. This qualitatively matches the reported visual bias of humans engaged in audio-visual tasks. Our results confirm that distinct configurations of feedforward/feedback connectivity have an important functional impact on a model's behavior. Therefore, top-down feedback captures behaviors and perceptual preferences that do not manifest reliably in feedforward-only networks. Further experiments are needed to

clarify whether top-down feedback helps an ANN fit better to neural data, but the results show that top-down feedback affects the processing of stimuli and is thus a relevant feature that should be considered for deep ANN models in computational neuroscience more broadly.”

(2) It is not clear what the advantages are between the brain-like model and a feedforward-only model in terms of performance in solving the task. Given Figures 4 and 5, it is evident that the feedforward-only model reaches almost the same performance as the brain-like model (when the latter uses the modulatory feedback with the composite function) on almost all tasks tested. The speed of learning is nearly the same: for some tested tasks the brain-like model learns faster, while for others it learns slower. Thus, it is hard to attribute a functional implication to the feedback connections given the presented figures and therefore the strong claims in the Discussion should be rephrased or toned down.

Again, we believe that there has been a misunderstanding regarding the goals of this study, as we are not trying to claim here that there are performance advantages conferred by top-down feedback in this case. Indeed, we share the reviewer’s assessment that the feedforward only model seems to be capable of solving this task well. To reiterate: our goal here was to demonstrate that top-down feedback alters the computations in the network and, thus, has distinct effects on behaviour that need to be considered by researchers who use deep networks to model the brain. But we make no claims of “superiority” of the brain-like model.

In-line with this, we’re not completely sure which claims in the discussion the reviewer is referring to. We note that we were quite careful in our claims. For example, in the first section of the discussion we say:

“Altogether, our results demonstrate that the distinction between feedforward and feedback inputs has clear computational implications, and that ANN models of the brain should therefore consider top-down feedback as an important biological feature.”

And later on:

“In summary, our study shows that modulatory top-down feedback and the architectural diversity enabled by it can have important functional implications for computational models of the brain. We believe that future work examining brain function with deep neural networks should therefore consider incorporating top-down modulatory feedback into model architectures when appropriate.”

If we have missed a claim in the discussion that implies superiority of the brain-like model in terms of task performance we would be happy to change it.

(3) The Methods section lacks sufficient detail. There is no explanation provided for the choice of hyperparameters nor for the structure of the networks (number of trainable parameters, number of nodes per layer, etc). Clarifying the rationale behind these decisions would enhance understanding. Moreover, since the authors draw conclusions based on the performance of the networks on specific tasks, it is unclear whether the comparisons are fair, particularly concerning the number of trainable parameters. Furthermore, it is not clear if the visual bias observed in the brain-like model is an emerging property of the network or has been created because of the asymmetries in the visual vs. auditory pathway (size of the layer, number of layers, etc).

We thank the reviewer for raising this issue, and want to provide some clarifications: First, the number of trainable parameters are roughly equal, since we were only switching the direction of connectivity (top-down versus bottom-up), not the number of connections. We confirmed the biggest difference in size is between models with composite and multiplicative

feedback; models with composite feedback have roughly ~1K more parameters, and all models are within the 280K parameter range. We now state this in the methods.

Second, because superior performance was not the goal of this study, as stated above, we conducted limited hyperparameter tuning. Given the reviewer's comment, we wondered whether this may have impacted our results. Therefore, we explored different hyperparameters for the model during the multimodal auditory tasks, which show the clearest example of the visual dominance in the brainlike model (Figure 3).

We explored different hidden state sizes, learning rates and processing times, and examined whether the core results were different. We found that extremely high learning rates (0.1) destabilize all models and that some models perform poorly under different processing times. But overall, the core results are evident across all hyperparameters where the models learn i.e the different behaviors of models with different connectivities and the visual dominance observed in the brainlike model. We now provide these results in a supplementary figure (Fig. S2, showing larger models trained with different learning rates, and Fig S3, which shows the effect of processing time on AS task performance).

Reviewer #2 (Public review):

Summary:

This work addresses the question of whether artificial deep neural network models of the brain could be improved by incorporating top-down feedback, inspired by the architecture of the neocortex.

In line with known biological features of cortical top-down feedback, the authors model such feedback connections with both, a typical driving effect and a purely modulatory effect on the activation of units in the network.

To assess the functional impact of these top-down connections, they compare different architectures of feedforward and feedback connections in a model that mimics the ventral visual and auditory pathways in the cortex on an audiovisual integration task.

Notably, one architecture is inspired by human anatomical data, where higher visual and auditory layers possess modulatory top-down connections to all lower-level layers of the same modality, and visual areas provide feedforward input to auditory layers, whereas auditory areas provide modulatory feedback to visual areas.

First, the authors find that this brain-like architecture imparts the models with a light visual bias similar to what is seen in human data, which is the opposite in a reversed architecture, where auditory areas provide a feedforward drive to the visual areas.

Second, they find that, in their model, modulatory feedback should be complemented by a driving component to enable effective audiovisual integration, similar to what is observed in neural data.

Last, they find that the brain-like architecture with modulatory feedback learns a bit faster in some audiovisual switching tasks compared to a feedforward-only model.

Overall, the study shows some possible functional implications when adding feedback connections in a deep artificial neural network that mimics some functional aspects of visual perception in humans.

Strengths:

The study contains innovative ideas, such as incorporating an anatomically inspired architecture into a deep ANN, and comparing its impact on a relevant task to alternative

architectures.

Moreover, the simplicity of the model allows it to draw conclusions on how features of the architecture and functional aspects of the top-down feedback affect the performance of the network.

This could be a helpful resource for future studies of the impact of top-down connections in deep artificial neural network models of the neocortex.

We thank the reviewer for their summary and their recognition of the innovative components and helpful resources therein.

Weaknesses:

Overall, the study appears to be a bit premature, as several parts need to be worked out more to support the claims of the paper and to increase its impact.

First, the functional implication of modulatory feedback is not really clear. The "only feedforward" model (is a drive-only model meant?) attains the same performance as the composite model (with modulatory feedback) on virtually all tasks tested, it just takes a bit longer to learn for some tasks, but then is also faster at others. It even reproduces the visual bias on the audiovisual switching task. Therefore, the claims "Altogether, our results demonstrate that the distinction between feedforward and feedback inputs has clear computational implications, and that ANN models of the brain should therefore consider top-down feedback as an important biological feature." and "More broadly, our work supports the conclusion that both the cellular neurophysiology and structure of feed-back inputs have critical functional implications that need to be considered by computational models of brain function" are not sufficiently supported by the results of the study. Moreover, the latter points would require showing that this model describes neural data better, e.g., by comparing representations in the model with and without top-down feedback to recorded neural activity.

To emphasize again our specific claims, we believe that our data shows that top-down feedback has functional implications for deep neural network behaviour, not increased performance or neural alignment. Indeed, our results demonstrate that top-down feedback alters the behaviour of the networks, as shown by the differences in responses to various combinations of ambiguous stimuli. We agree with the reviewer that if our goal was to claim either superior performance on these tasks, or better fit to neural data, we would need to actually provide data supporting that claim.

Given the comments from the reviewer, we have tried to provide more clarity in the introduction and discussion regarding our claims. In particular, we now highlight that we are not trying to demonstrate that the models with top-down feedback exhibit superior performance or better fit to neural data.

As one final note, yes, the reviewer understood correctly that the "only feedforward" model is a model with only driving inputs. We have renamed the feedforward-only models to drive only models and added additional emphasis in the text to ensure that the distinction is clear for all readers.

Second, the analyses are not supported by supplementary material, hence it is difficult to evaluate parts of the claims. For example, it would be helpful to investigate the impact of the process time after which the output is taken for evaluation of the model. This is especially important because in recurrent and feedback models the convergence should be checked, and if the network does not converge, then it should be discussed why at which point in time the network is evaluated.

This is an excellent point, and we thank the reviewer for raising it. We allowed the network to process the stimuli for seven time-steps, which was enough for information from any one region to be transmitted to any other. We found in some initial investigations that if we shortened the processing time some seeds would fail to solve the task. But, based on the reviewer's comment, we have now also run additional tests with longer processing times for the auditory tasks where we see the clearest visual bias (Figure 3). We find that different process times do not change the behavioral biases observed in our models, but may introduce difficulties ignoring visual stimuli for some models. Thus, while process time is an important hyperparameter for optimal performance of the model, the central claim of the paper remains. We include this new data in a supplementary figure S3.

Third, the descriptions of the models in the methods are hard to understand, i.e., parameters are not described and equations are explained by referring to multiple other studies. Since the implications of the results heavily rely on the model, a more detailed description of the model seems necessary.

We agree with the reviewer that the methods could have been more thorough. Therefore, we have greatly expanded the methods section. We hope the model details are now more clear.

Lastly, the discussion and testable predictions are not very well worked out and need more details. For example, the point "This represents another testable prediction flowing from our study, which could be studied in humans by examining the optical flow (Pines et al., 2023) between auditory and visual regions during an audiovisual task" needs to be made more precise to be useful as a prediction. What did the model predict in terms of "optic flow", how can modulatory from simple driving effect be distinguished, etc.

We see that the original wording of this prediction was ambiguous, thank you for pointing this out. In the study highlighted (Pines et al., 2023) the authors use an analysis technique for measuring information flow between brain regions, which is related to analysis of optical flow in images, but applied to fMRI scans. This is confusing given the current study, though. Therefore, we have changed this sentence to make clear that we are speaking of information flow here.

Reviewer #3 (Public review):

Summary:

This study investigates the computational role of top-down feedback in artificial neural networks (ANNs), a feature that is prevalent in the brain but largely absent in standard ANN architectures. The authors construct hierarchical recurrent ANN models that incorporate key properties of top-down feedback in the neocortex. Using these models in an audiovisual integration task, they find that hierarchical structures introduce a mild visual bias, akin to that observed in human perception, not always compromising task performance.

Strengths:

The study investigates a relevant and current topic of considering top-down feedback in deep neural networks. In designing their brain-like model, they use neurophysiological data, such as externopyramidalisation and hierarchical connectivity. Their brain-like model exhibits a visual bias that qualitatively matches human perception.

We thank the reviewer for their summary and evaluation of our paper's strengths.

Weaknesses:

While the model is brain-inspired, it has limited bioplausibility. The model assumes a simplified and fixed hierarchy. In the brain with additional neuromodulation, the hierarchy could be more flexible and more task-dependent.

We agree, there are still many facets of top-down feedback that we have not captured here, and the modulation of hierarchy is an interesting example. We have added some consideration of this point to the limitations section of the discussion.

While the brain-like model showed an advantage in ignoring distracting auditory inputs, it struggled when visual information had to be ignored. This suggests that its rigid bias toward visual processing could make it less adaptive in tasks requiring flexible multimodal integration. It hence does not necessarily constitute an improvement over existing ANNs. It is unclear, whether this aspect of the model also matches human data. In general, there is no direct comparison to human data. The study does not evaluate whether the top-down feedback architecture scales well to more complex problems or larger datasets. The model is not well enough specified in the methods and some definitions are missing.

We agree with the reviewer that we have not demonstrated anything like superior performance (since the brain-like network is quite rigid, as noted) nor have we shown better match to human data with the brain-like network. This was not our intended claim. Rather, we demonstrated here simply that top-down feedback impacts behavior of the networks in response to ambiguous stimuli. We have now added statements to the introduction and discussion to make our specific claims (which are supported by our data, we believe) clear.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

I believe that the work is very nice but not so mature at this stage. Below, you can find some comments that eventually could improve your manuscript.

(1) Intro, last sentence: "Therefore, top-down feedback is a relevant feature that should be considered for deep ANN models in computational neuroscience more broadly." I don't understand what the authors refer to with this sentence. There are numerous models (deep ANNs) that have been used to model the neural activity and are much simpler than the one proposed here which contains very complex models and connectivity. Although I do agree that the top-down connections are very important there is no data to support their importance for modeling the brain.

Respectfully, we disagree with the reviewer that we don't provide data to demonstrate the importance of top-down feedback for modelling. Indeed, we provided a great deal of data to show that top-down feedback in the networks has real functional implications for behaviour, e.g., it can induce a human-like visual bias. Thus, top-down feedback is a factor that one should care about when modelling the brain. But, we agree with the reviewer that more demonstration of the utility of using top-down feedback for achieving better fits to neural data would be an important next step.

(2) I suggest adding some extra supplementary simulations where, for example, the number of data for visual and auditory pathways is equal in size (i.e., the same number of examples), the number of layers is identical (3 per pathway), and also the number of parameters. Doing this would help strengthen the claims presented in the paper.

In fact, all of the hyperparameters the reviewer mentions here were identical for the different networks, so the experiments the reviewer is requesting here were already part of the paper. We now clarify this in the text.

(3) Results: I suggest adding Tables with quantifications of the presented results. For example, best performance, epochs to converge, etc. As it is now, it is very hard to follow the evidence shown in Figures.

This is a good suggestion, we have now added this table to the start of the supplemental figures.

(4) Figure 2e, 3e: Although VS3, and AS3 have been used only for testing, the plot shows alignments with respect to training epochs. The authors should clarify in the Methods if they tested the network with all intermediate weights during VS1/VS2 or AS1/AS2 training.

Testing scenarios in this context meant that the model was never shown the scenario/task during training, but the models were indeed evaluated on the VS3 and AS3 after each training epoch. We have added clarifications to the figure legends.

(5) Methods: It would be beneficial to discuss how specific hyperparameters were selected based on prior research, empirical testing, or theoretical considerations. Also, it is not clear how the alignment (visual or audio) is calculated. Do the authors use the examples that have been classified correctly for both stimuli or do they exclude those from the analysis (maybe I have missed it).

As noted above, because superior performance was not the goal of this study, we conducted limited hyperparameter tuning. But we have extended the results with additional hyperparameter tuning in a supplementary figure, and describe the hyperparameter choices more thoroughly in the methods. As well, all data includes all model responses, regardless of whether they were correct or not. We now clarify this in the methods.

(6) Code: The code repository lacks straightforward examples demonstrating how to utilize the modeling approach. Given that it is referred to as a "framework", one would expect it to facilitate easy integration into various models and tasks. Including detailed instructions or clear examples would significantly improve usability and help users effectively apply the proposed methodology.

We agree with the reviewer, this would be beneficial. We have revised the README of the codebase to explain the model and its usage more clearly and included an interactive jupyter notebook with example training on MNIST.

Some minor comments are given below. Generally speaking, the Figures need to be more carefully checked for consistent labels, colors, etc.

(1) Page 4, 1st paragraph - grammar correction: "a larger infragranular layer" or "larger infragranular layers"

Thank you for catching this, we have fixed the text.

(2) Page 4, 2nd para - rephrase: "In three additional control ANNs" → "In the third additional control ANN"

In fact, we did mean three additional control ANNs, each one representing a different randomized connectivity profile. We now clarify this in the text and provide the connectivity of the two other random graphs in the supplemental figures.

(3) Page 4, VAE acronym needs to be defined before its first use

The variational autoencoder is introduced by its full name in the text now.

(4) Page 4: Fig. 2c reference should be Fig. 2b, Fig. 2d should be Fig. 2c, Fig. 2b should be Fig. 2d, VS4; Fig. 2b, bottom should be VS4; Fig. 2f, Fig. 2f to Fig. 2g. Double check the Figure references in the text. Here is very confusing for the reader.

We have now fixed this, thank you for catching it.

(5) Page 5, 1st para: "Altogether, our results demonstrated both" → "Altogether, our results demonstrated that both"

This has been updated.

(6) Figure 2: In the e and g panels the x label is missing.

This was actually because the x-axis were the same across the panels, but we see how this was unclear, so we have updated the figure.

(7) Figure 3: There is no panel g (the title is missing); In panels b, c, e, and g the y label is missing, and in panels e and g the x label is missing. Also, the Feedforward model is shown in panel g but it is introduced later in the text. Please remove it from Figure 3. Also in legend: "AV Reverse graph" → "Reverse graph". Also, "Accuracy" and "Alignment" should be presented as percentages (as in Figure 2).

This has been corrected.

(8) Figure 4; x labels are missing.

As with point (6), this was actually because the x-axis were the same across the panels, but we see how this was unclear, so we have updated the figure.

(9) Page 7; I can't find the cited Figure S1.

Apologies, we have added the supplemental figure (now as S4). It shows the results of models with multiplicative feedback on the task in Fig 5 (as opposed to models with composite feedback shown in the main figure).

Reviewer #2 (Recommendations for the authors):

(1) Discussion Section 3.1 is only a literature review, and does not really add any value.

Respectfully, we think it is important to relate our work to other computational work on the role of top-down feedback, and to make clear what our specific contribution is. But, we have updated the text to try to place additional emphasis on our study's contribution, so that this section is more than just a literature review.

"Our study adds to this previous work by incorporating modulatory top-down feedback into deep, convolutional, recurrent networks that can be matched to real brain anatomy. Importantly, using this framework we could demonstrate that the specific architecture of top-down feedback in a neural network has important computational implications, endowing networks with different inductive biases."

(2) Including ipython notebooks and some examples would be great to make it easier to use the code.

We now provide a demo of how to use the code base in a jupyter notebook.

(3) The description of the model is hard to comprehend. Please name and describe all parameters. Also, a figure would be great to understand the different model equations.

We have added definitions of all model terms and parameters.

(4) The terminology is not really clear to me. For example "The results further suggest that different configurations of top-down feedback make otherwise identically connected models functionally distinct from each other and from traditional feedforward only recurrent models." The feedforward and only recurrent seem to contradict each other. Would maybe driving and modulatory be a better term here? I also saw in the code that you differentiate between three types of inputs, modulatory, threshold offset and basal (like feedforward). How about you only classify connections based on these three type? I was also confused about the feedforward only model, because I was unsure whether it is still feedback connections but with "basal" quality, or whether feedback connections between modalities and higher-to-lower level layers were omitted altogether.

We take the reviewer's point here. To clarify this, we have updated the text to refer to "driving only" rather than "feedforward only", to make it obvious that what we change in these models is simply whether the connection has any modulatory impact on the activity.

(5) "incorporating it into ANNs can affect their behavior and help determine the solutions that the network can discover." -> Do you mean constrain? Overall, I did not really get this point.

Yes, we mean that it constrains the solutions that the network is likely to discover.

(6) "ignore the auditory inputs when they visual inputs were unambiguous" -> the not they

This has been fixed. Thank you for catching it.

(7) xlabel in Figure 4 is missing.

This has been fixed, thank you for catching it.

Reviewer #3 (Recommendations for the authors):

Major:

(1) How alignment is computed is not defined. In addition to a proper definition in the methods section, it would be nice to briefly define it when it first appears in the results section.

We've added an explicit definition of how alignment is calculated in the methods and emphasized the calculation when its first explained in the results

(2) A connectivity matrix for the feedforward-only model is missing and could be added.

We have added this to Figure 1.

(3) The connectivity matrix for each random model should also be shown.

We've shown each of the random model configurations in the new supplemental figure S1.

(4) Initial parameters are not defined, such as W , b etc. A table with all model parameters would be great.

We have added a table to the methods listing all of the parameters.

(5) Would be nice to show the t-sne plots (not just the NH score) for each model and each task in the appendix.

We can provide these figures on request. They massively increase the file size of the paper pdf, as there's 49 of them for each task and each model, 980 in total. An example t-SNE plot is

provided in figure 6.

Minor:

(1) Page 4:

"we refer to this as Visual-dominant Stimulus case 1, or VS1; Fig. 1a, top)." This should be Fig. 2a.

(2) "In stimulus condition VS1, all of the models were able to learn to use the auditory clues to disambiguate the images (Fig. 2c)."

This should be Fig. 2b.

(3) "In comparison, in VS2, we found that the brainlike model learned to ignore distracting audio inputs quickly and consistently compared to the random models, and a bit more rapidly than the auditory information (Fig 2d)."

This should be Fig. 2c.

(4) "VS3; Fig. 2b, top"

This should be Fig. 2d

(5) "while all other models had to learn to do so further along in training (Fig. 2e)."

It is not stated explicitly, but this suggests that the image-aligned target was considered correct, and that weight updates were happening.

(6) "VS4; Fig. 2b, bottom"

This should be Fig. 2f

(7) "adept at learning (Fig. 2f)."

This should be Fig. 2g

(8) Figure 3:b,c,e y-labels are missing

3f: both x and y labels are missing

(9) Figure labeling in the text is not consistent (Fig. 1A versus Fig. 2a)

(10) Doubled "the" in ""This shows that the inductive bias towards vision in the brainlike model depended on the presence of the multiplicative component of the the feedback"

(11) Page 9 Figure 6: The caption says b shows the latent spaces for the VS2 task, whereas the main text refers to 6b as showing the latent space for the AS2 task. Please correct which task it is.

(12) Methods 4.1 page 13

"which is derived from the feedback input (h_{l-1})"

This should be h_{l+1}

(13) r_l , u_l , u and c are not defined to which aspects of the model they refer to

Even though this is based on a previous model, the methods section should completely describe the model.

Equations 1,2,3: the notation $[x;y]$ is unclear and should be defined.

Equation 5: u should probably be u_l .

(14) Page 14 typo: externopyrmidisation.

(15) It is confusing to use different names for the same thing: the all-feedforward model, the all feedforward network, the feedforward network, and the feedforward-only model are probably all the same? Consistent naming would help here.

Thank you for the detailed comments! We've fixed the minor errors and renamed the feedforward models to drive-only models.

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